

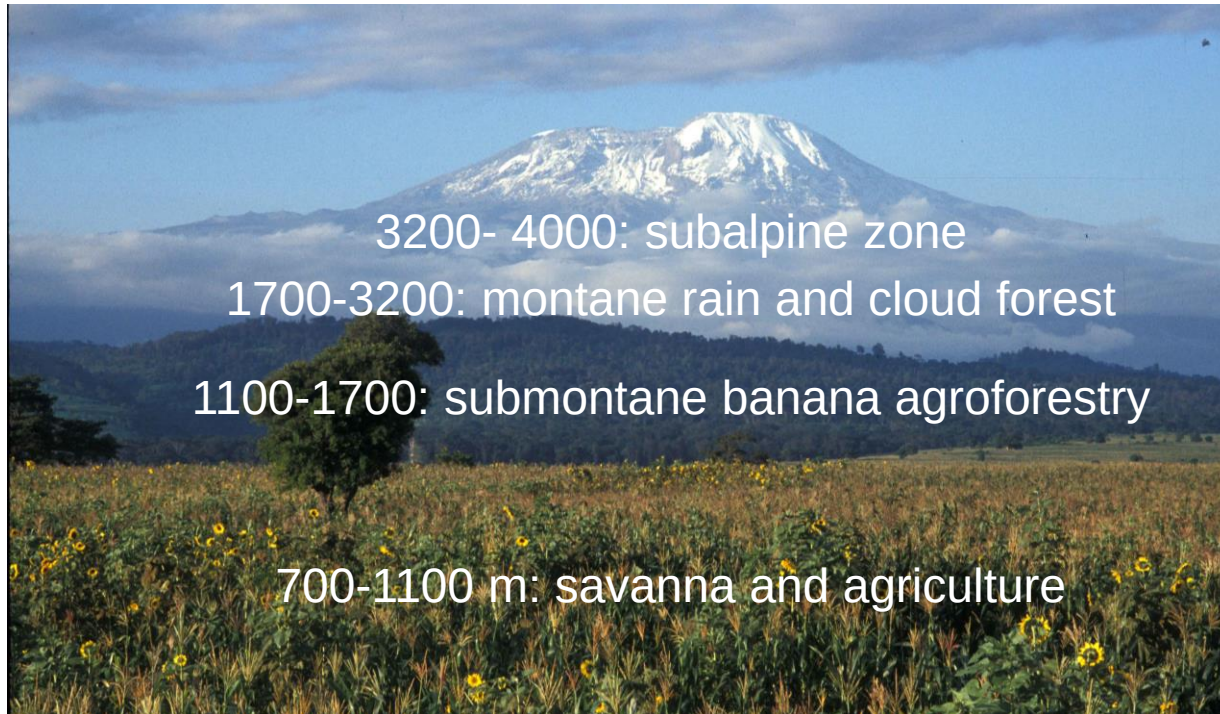
Environmental drivers of health in the Kilimanjaro Region, Tanzania

Netra Bhandari, Damiano Cerrone, Dirk Zeuss, Thomas Nauss
Environmental Informatics, Department of Geography

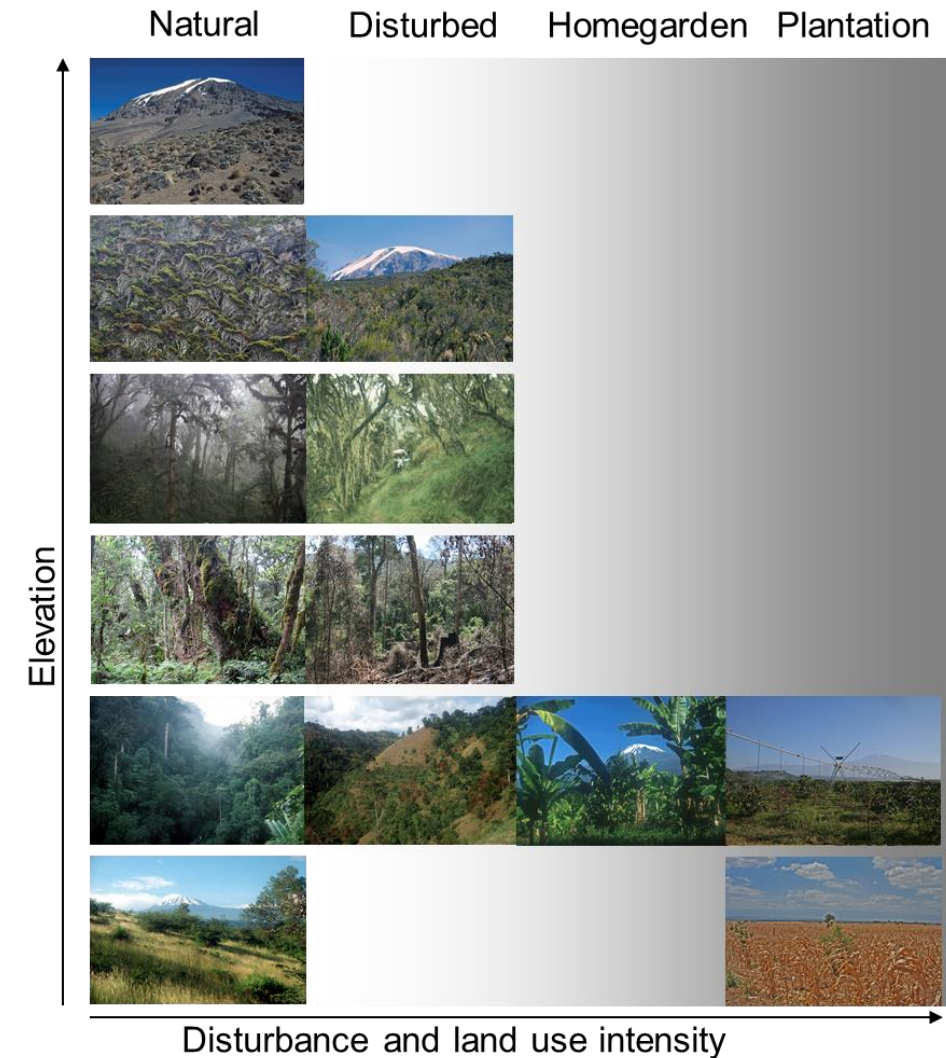
with all members of the RIPA2TAN project



Biodiversity, landuse, and people



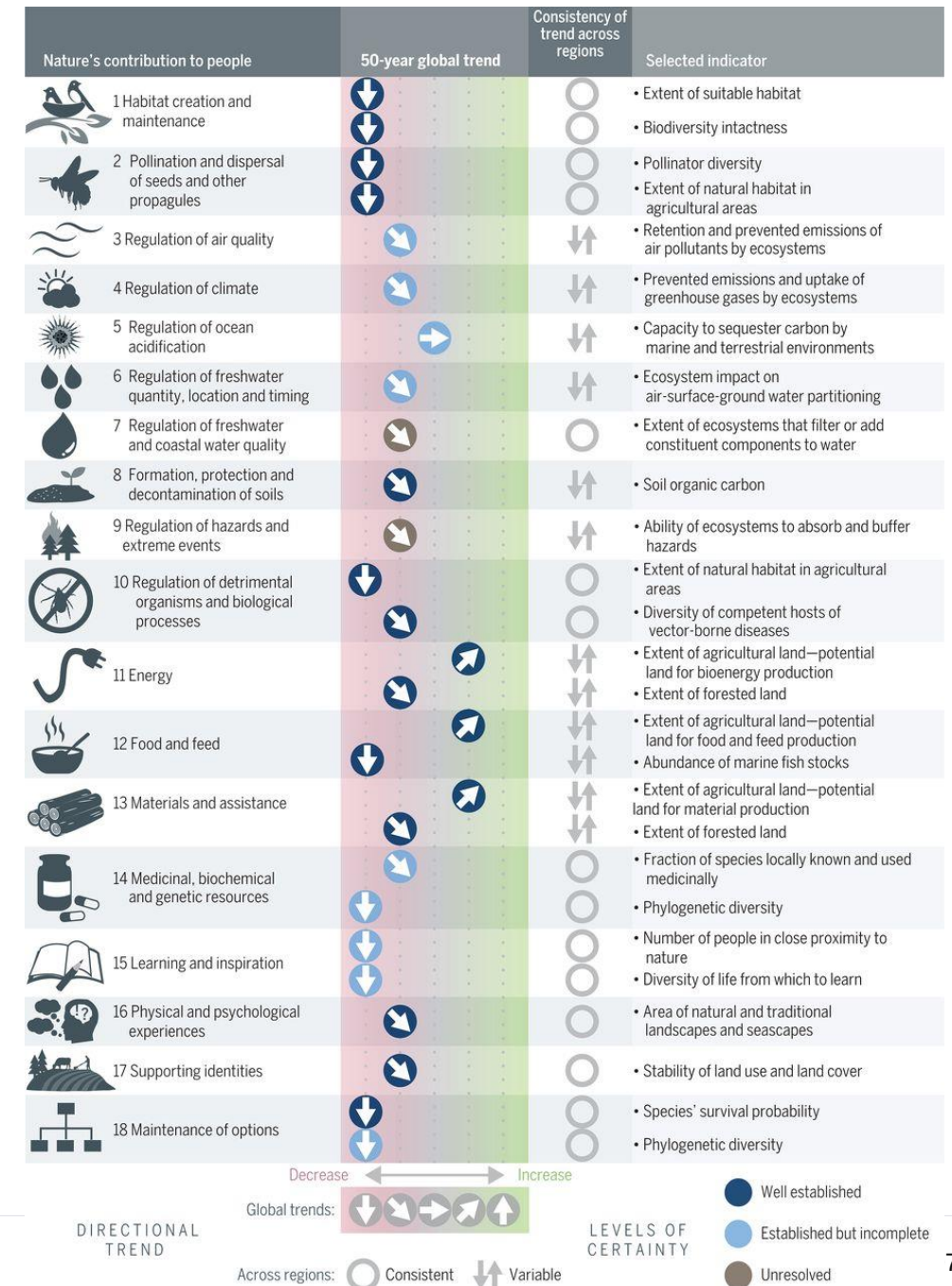
DFG project Kilimanjaro – Socioecological systems
<https://kili-ses.senckenberg.de/>



Nature's Contribution to People

Nature's contributions to people (NCP) are all the contributions, both positive and negative, of living nature (i.e. diversity of organisms, ecosystems, and their associated ecological and evolutionary processes) to the quality of life for people.

– Diaz et al 2018



Environment and health

- Hygiene hypothesis
- Biodiversity Hygiene hypothesis
- (von Hertzen et al 2011)

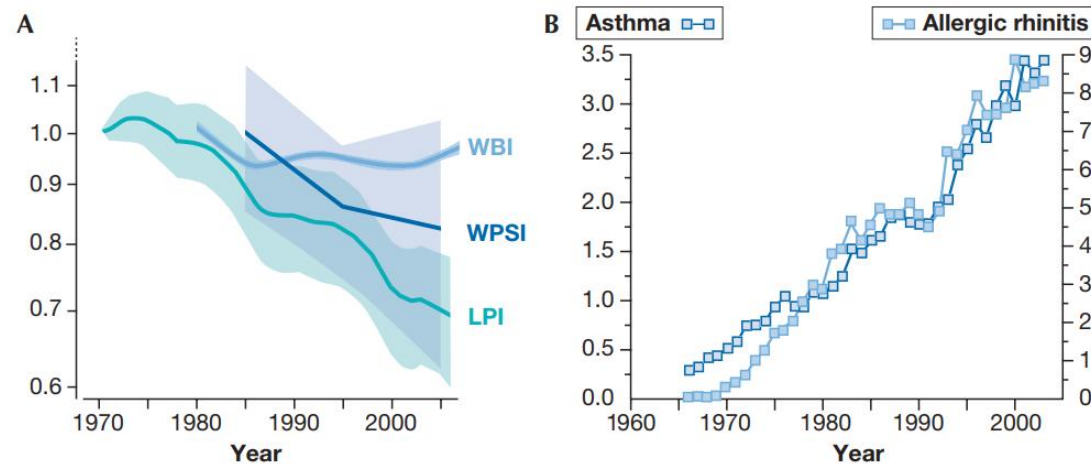
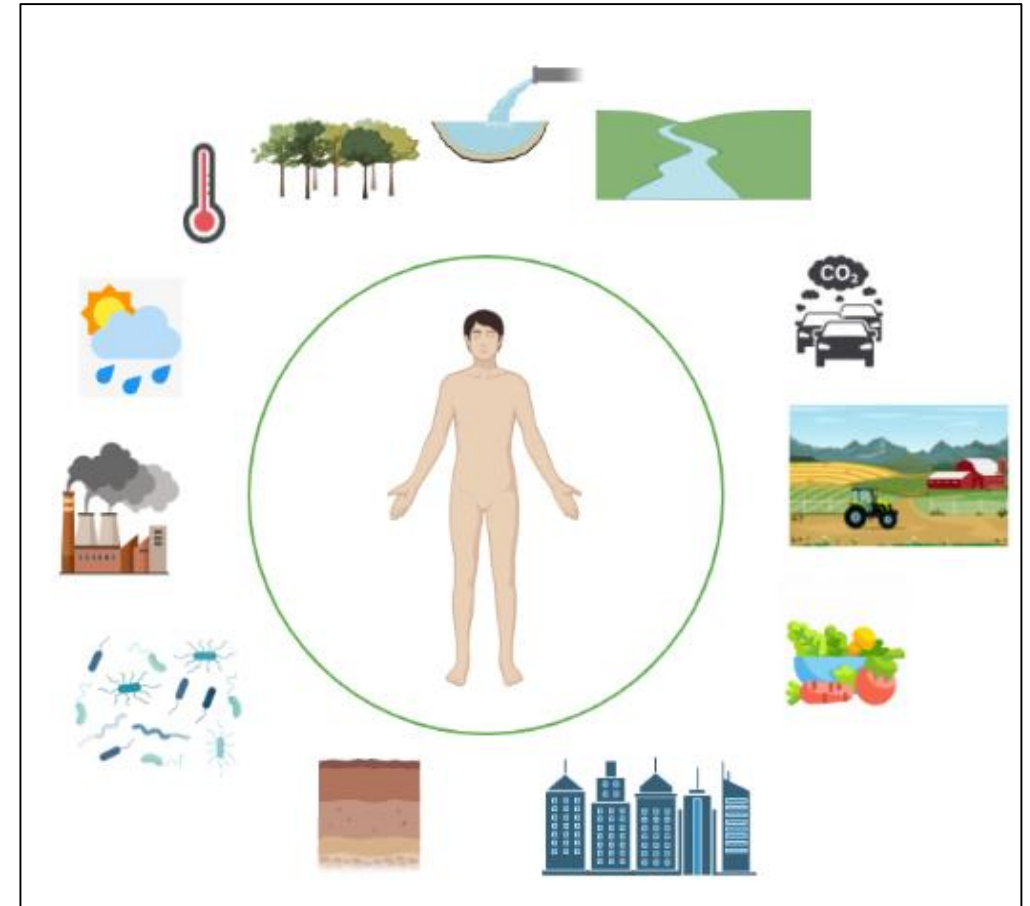


Fig 1 | Two global megatrends in biodiversity and public health. (A) Declining biodiversity since 1970 as measured by three indices. LPI, Living Planet Index; WBI, World Bird Index; WPSI, Waterbird Population Status Index (Butchart *et al*, 2010). (B) Increasing trends in the prevalence of inflammatory diseases. Asthma and allergic rhinitis among military conscripts from 1966 to 2003 (Latvala *et al*, 2005) are shown as an example.

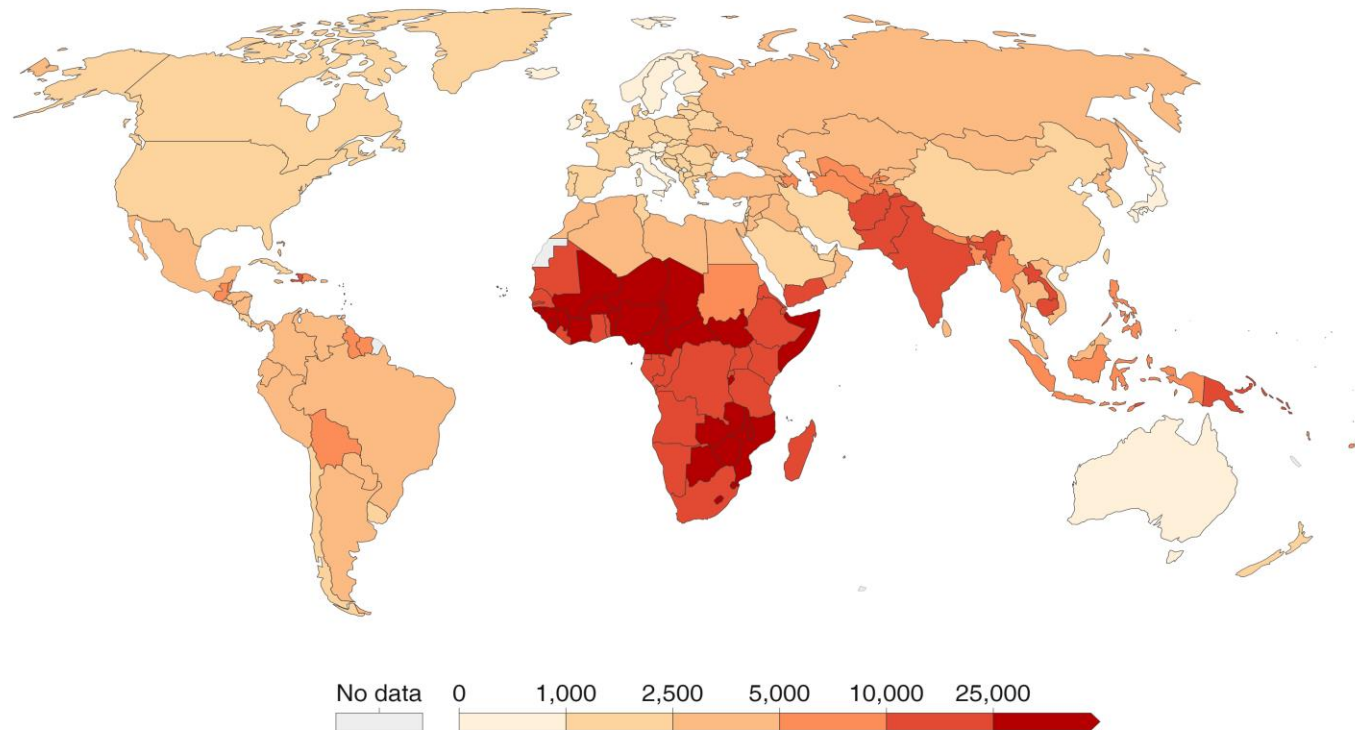


Global Burden of diseases- Communicable, neonatal, maternal & nutritional diseases

DALY rates from communicable, neonatal, maternal & nutritional diseases, 2019

Our World
in Data

Age-standardized DALY (Disability-Adjusted Life Year) rates per 100,000 individuals from communicable diseases. DALYs are used to measure total burden of disease - both from years of life lost and years lived with a disability. One DALY equals one lost year of healthy life.



Source: IHME, Global Burden of Disease

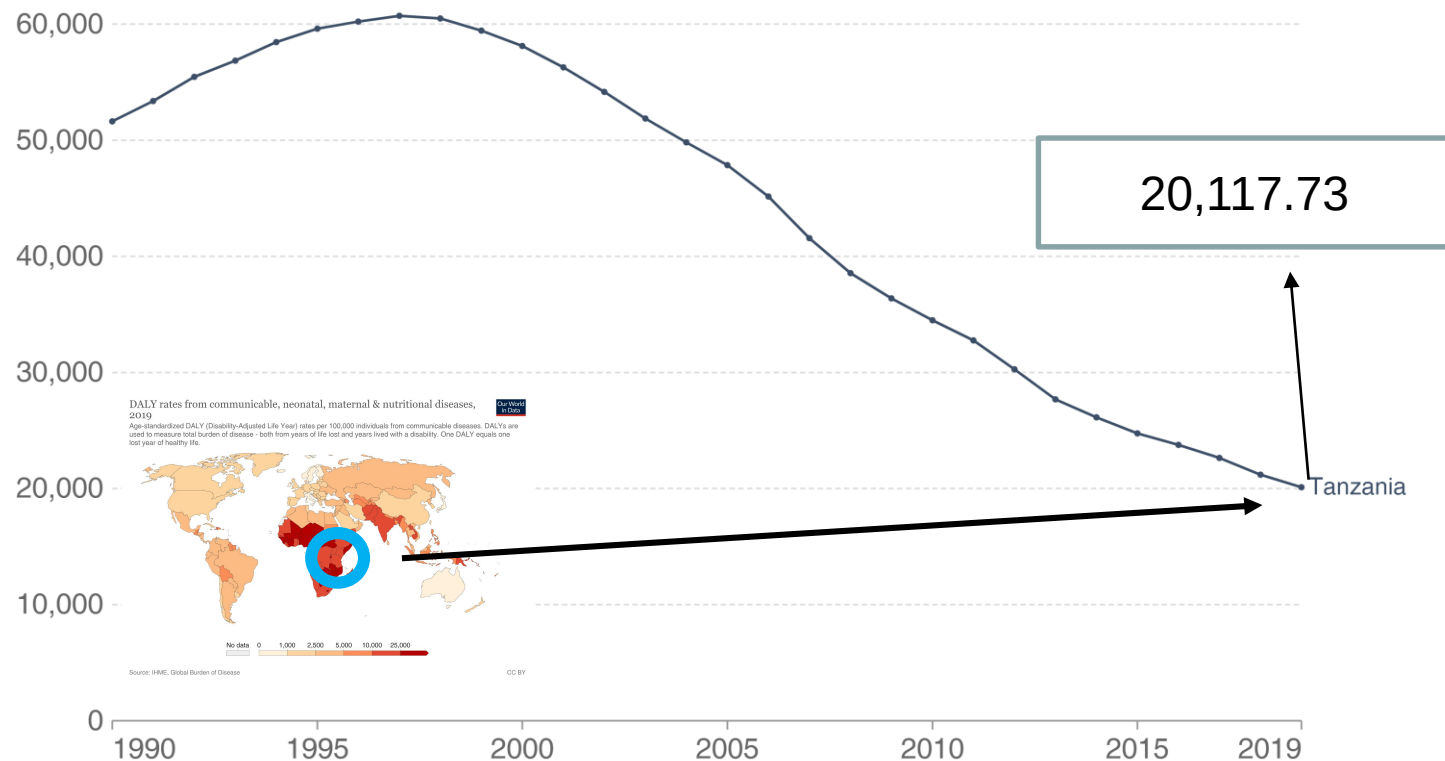
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Communicable, neonatal, maternal & nutritional diseases

DALY rates from communicable, neonatal, maternal & nutritional diseases, 1990 to 2019

Our World
in Data

Age-standardized DALY (Disability-Adjusted Life Year) rates per 100,000 individuals from communicable diseases. DALYs are used to measure total burden of disease - both from years of life lost and years lived with a disability. One DALY equals one lost year of healthy life.



Source: IHME, Global Burden of Disease

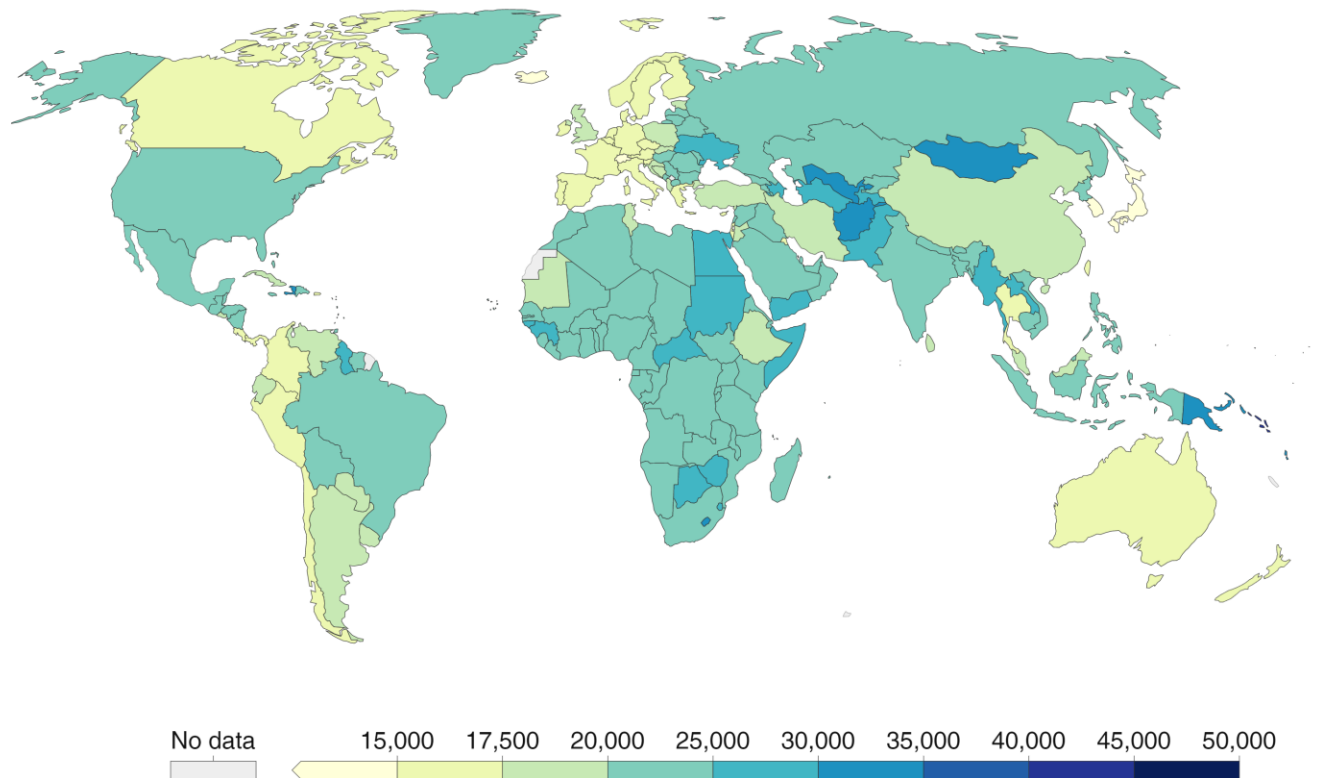
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Global burden of Non-communicable Diseases

DALY rates from non-communicable diseases (NCDs), 2019

Age-standardized DALY (Disability-Adjusted Life Year) rates per 100,000 individuals from non-communicable diseases (NCDs). DALYs are used to measure total burden of disease - both from years of life lost and years lived with a disability. One DALY equals one lost year of healthy life.

Our World
in Data



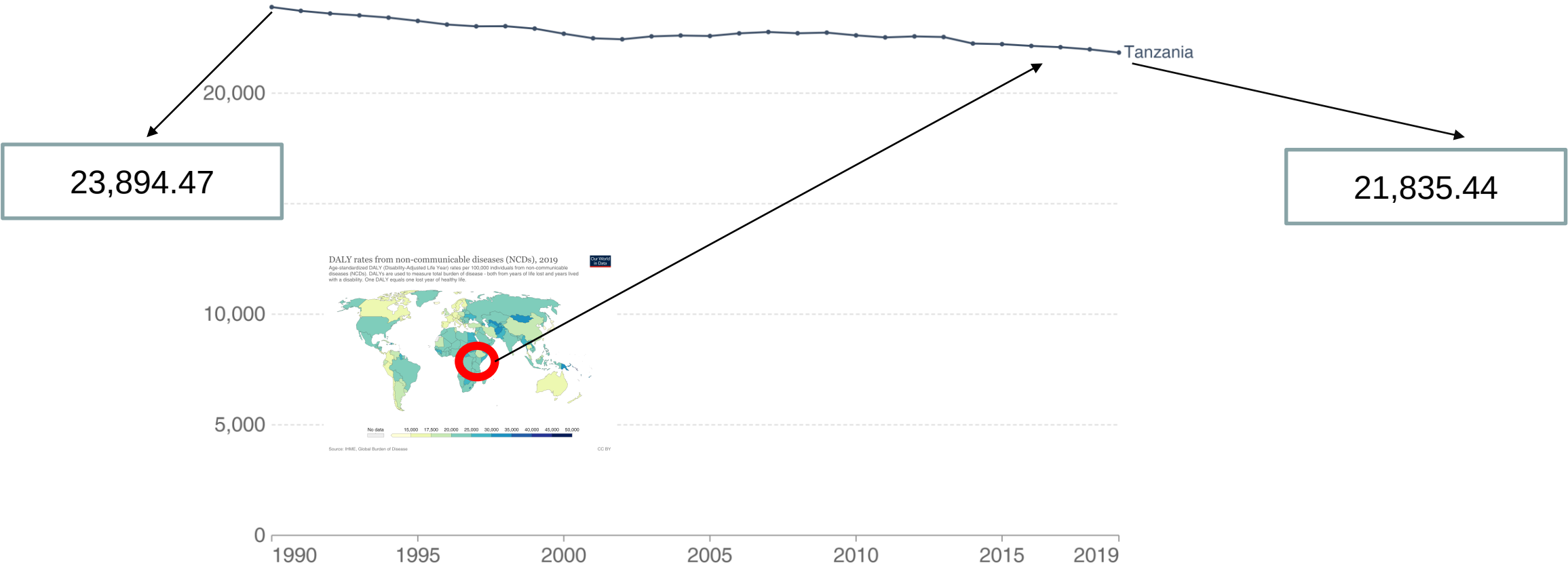
Source: IHME, Global Burden of Disease

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Non communicable diseases in Tanzania

DALY rates from non-communicable diseases (NCDs), 1990 to 2019

Age-standardized DALY (Disability-Adjusted Life Year) rates per 100,000 individuals from non-communicable diseases (NCDs). DALYs are used to measure total burden of disease - both from years of life lost and years lived with a disability. One DALY equals one lost year of healthy life.



Work packages

- WP1

Investigating the association of environmental and socioeconomic drivers with different **health indicators** in the Kilimanjaro region

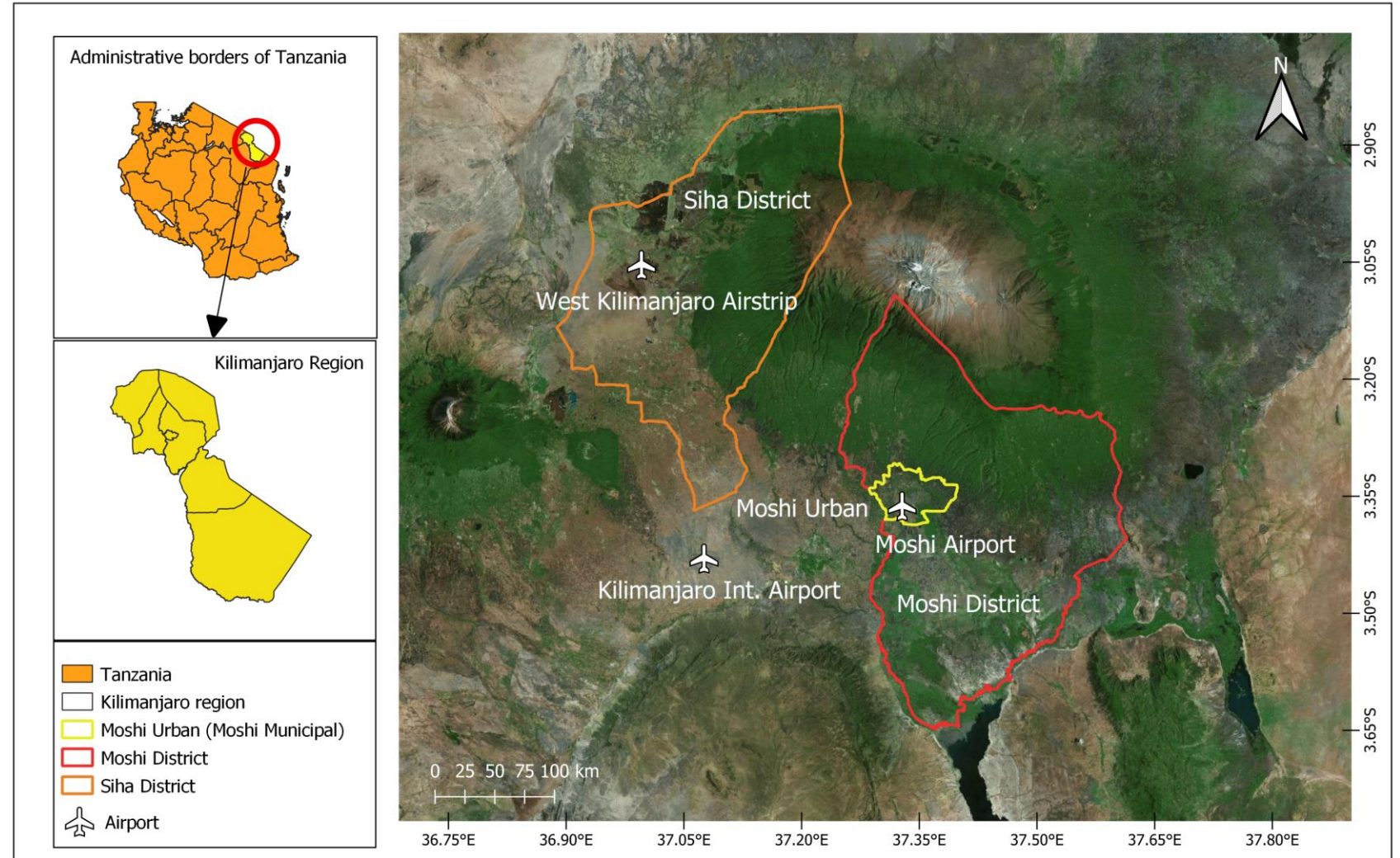
- WP2

Environmental drivers of **Asthma** in the Kilimanjaro region

Kilimanjaro region

Focus areas

- Siha District
- Moshi Urban



WP1

Objectives

- Does the percentage of communicable and non-communicable diseases differ between an urban and a rural area (/semi-urban)
- and are those differences driven by socioeconomic and/or environmental drivers

Status

Years in focus – 2018, 2019, 2020 (and 2021)

District-level data collection – Ongoing

Environmental variables - Ongoing

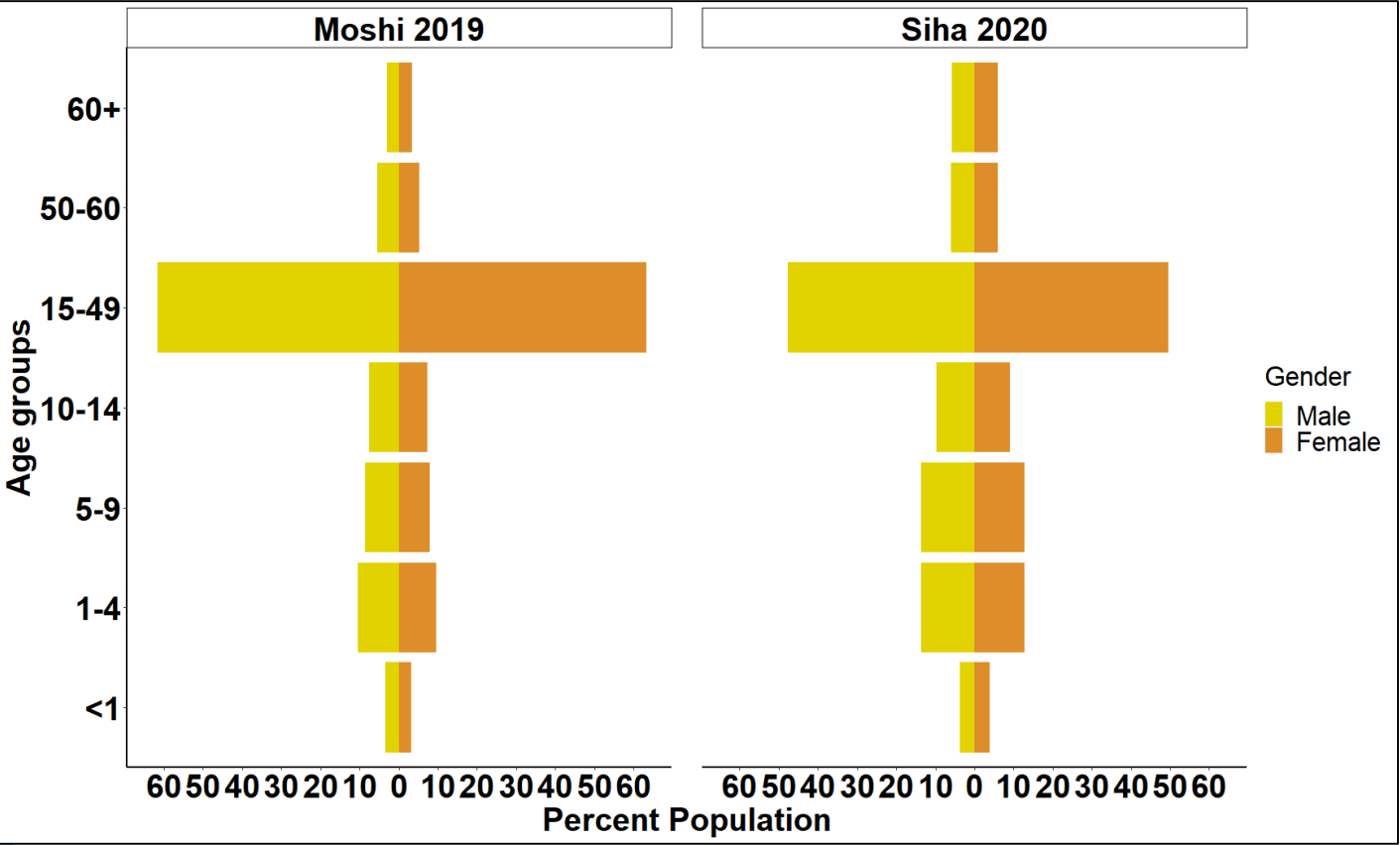
Methods and analysis – Ongoing

The following graphs are all based on currently available data

Moshi urban – 2019

Siha district - 2020

Demographics



District	Year	Total Population
Moshi Urban	2019	230,784
Siha	2020	141,568

Health indicators in the study

Health indicators	Moshi Urban (2019)	Siha (2020)
Birth rate (per 1000)		2.2
Neonatal mortality rate(per 1000)	230	3
Maternal mortality rate(per 1000)	31	154.75
Average life expectancy (years)	65	61
Access and use of medication/ prescribed medication (percent)	100%	

Socio economic variables

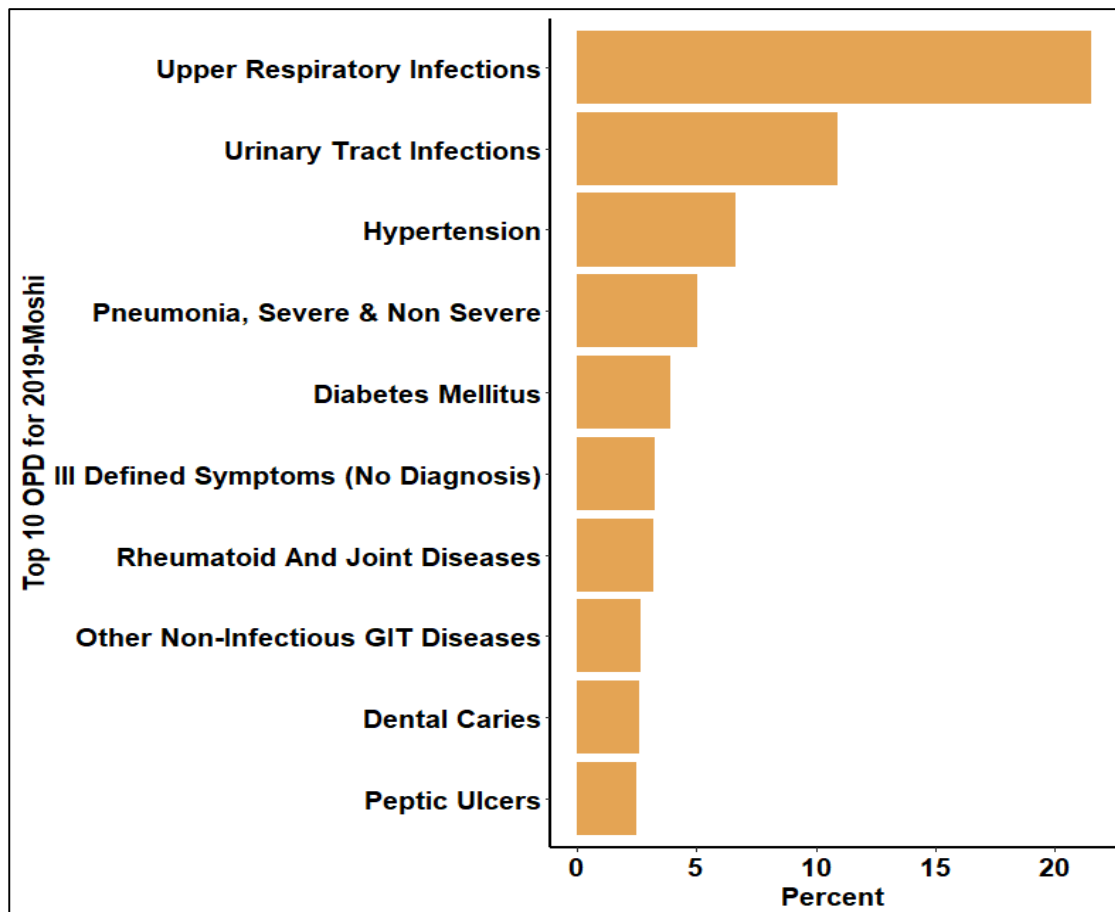
Socioeconomic data	Moshi Urban (2019)	Siha (2020)
Number of households with refrigerators		8848
Number of households with TVs or similar		26544
Household income		2504
Number of cars		
Number of motorbikes		
Unemployment rate		
People per household		4
Cellular phones		703448
No. of registered sim cards		

Healthcare access variables

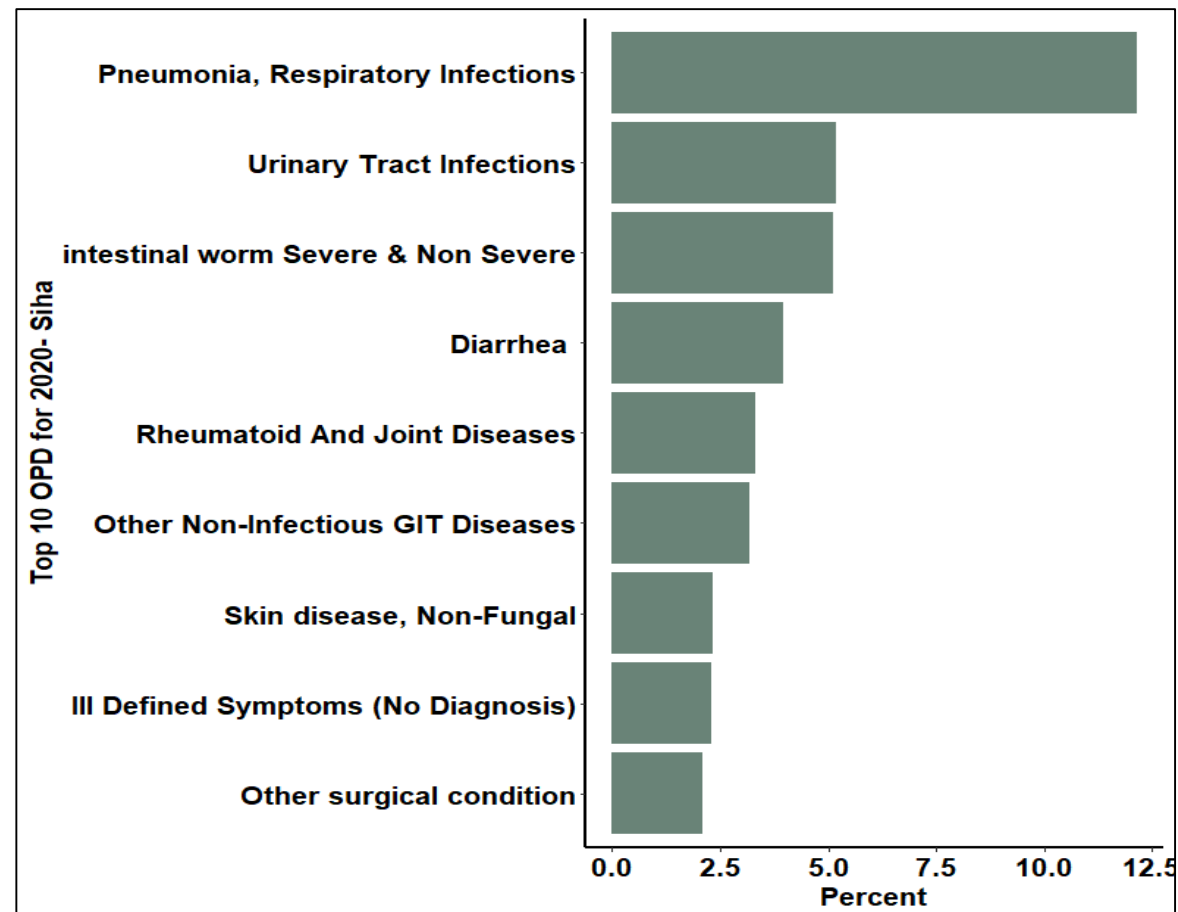
Healthcare access	Moshi Urban (2019)	Siha (2020)
Number of pharmacies	15	1
Number of hospitals	5	3
Number of hospital beds	220257	450
Number of doctors (Bachelor degree, Masters degree)		9
Number of clinical officers		24

Top 10 OPD – Outpatient department

Moshi Urban

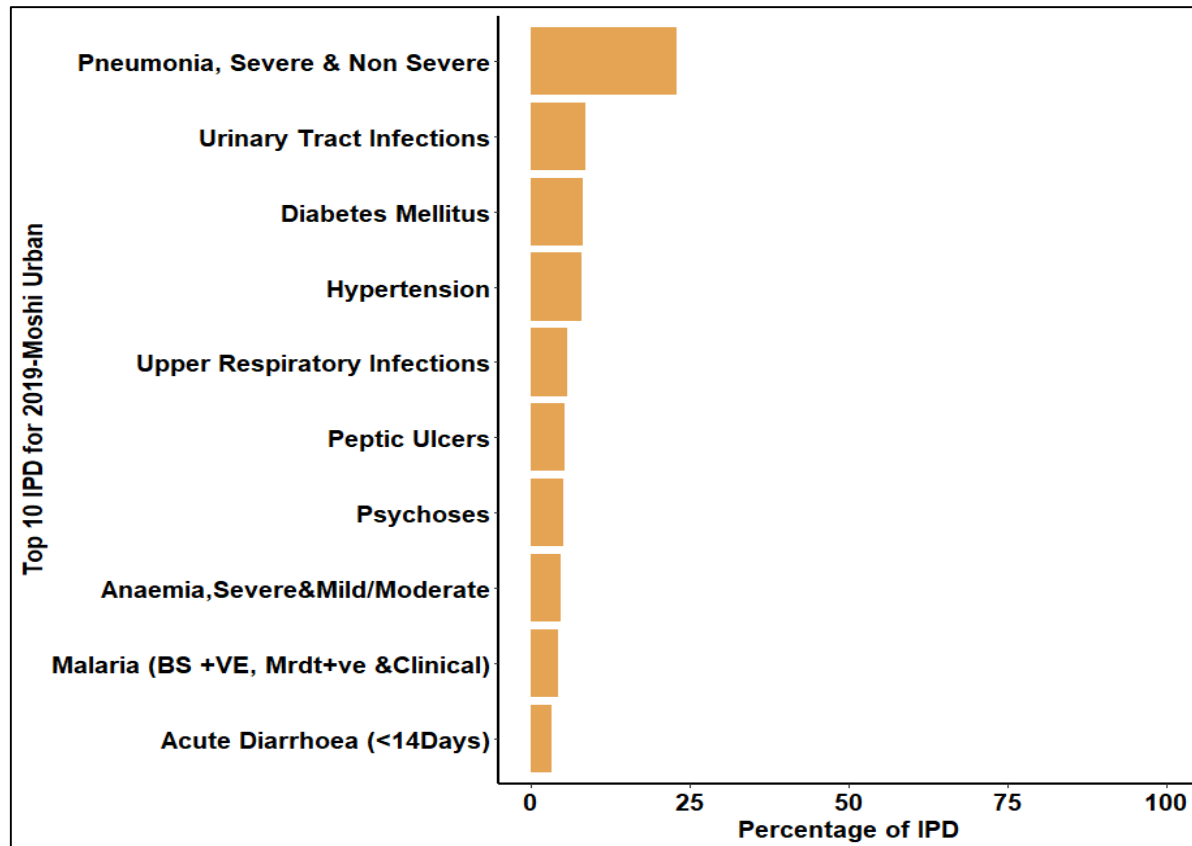


Siha

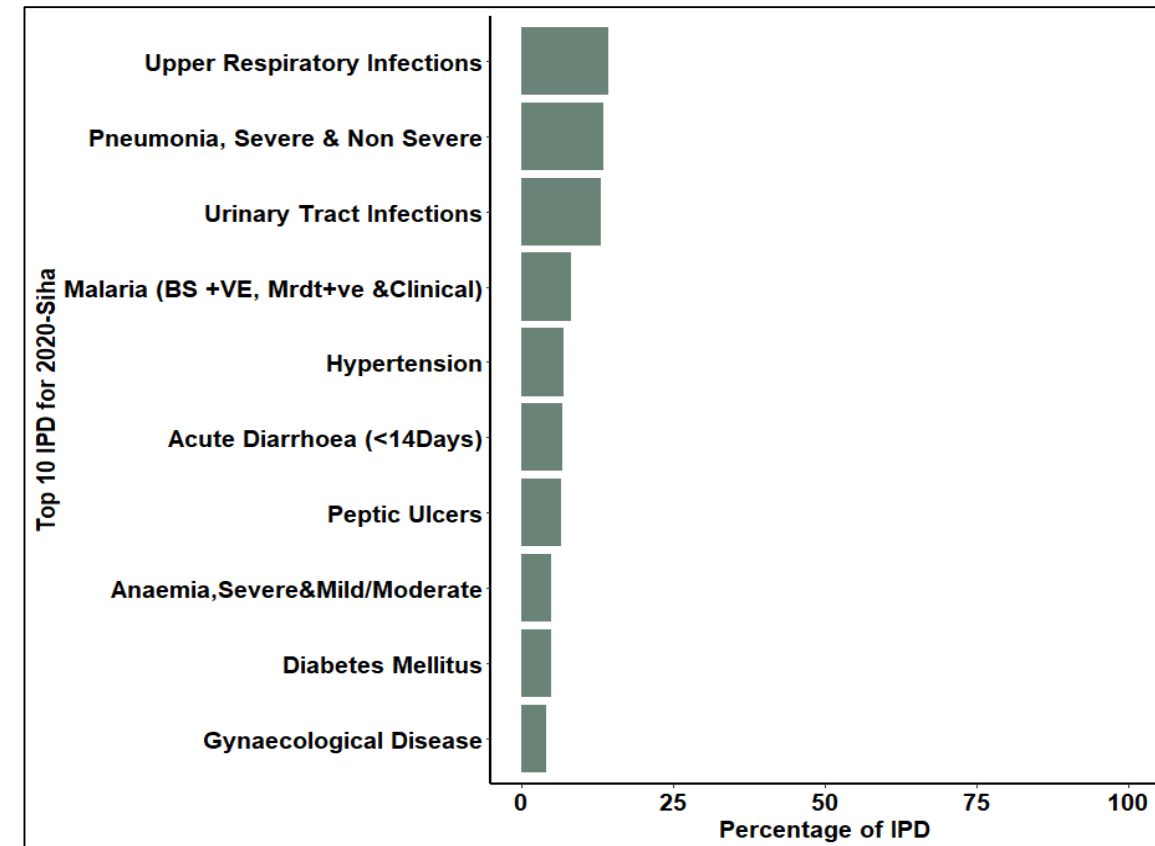


Top 10 IPD- inpatient department

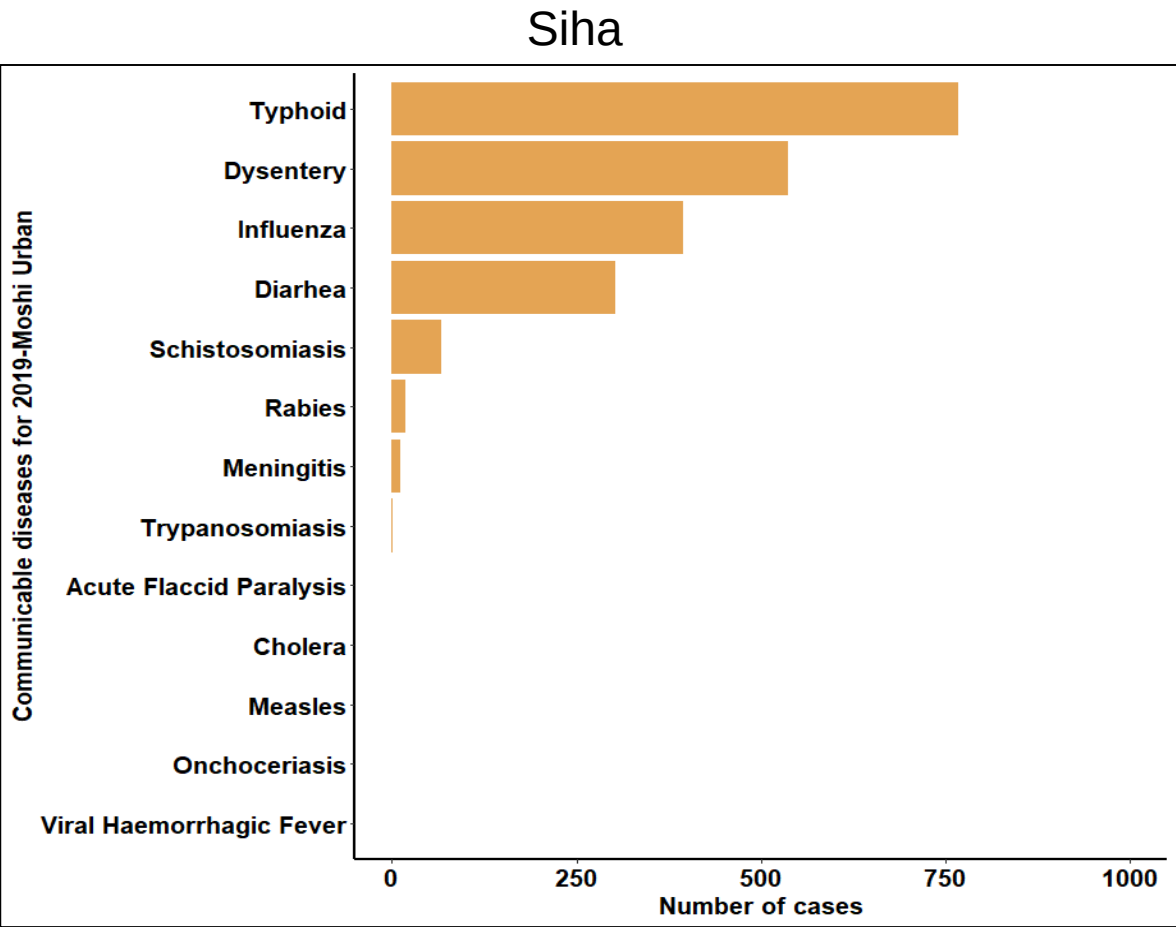
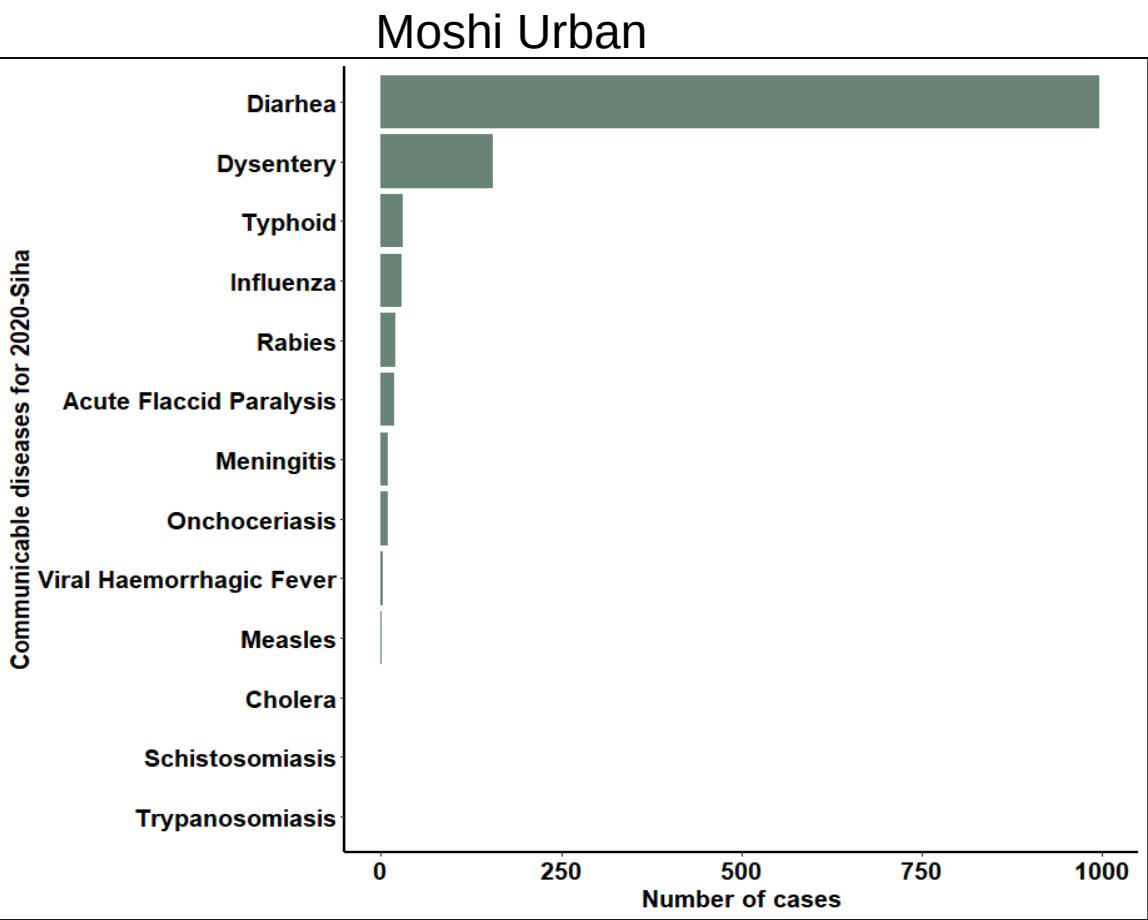
Moshi Urban



Siha

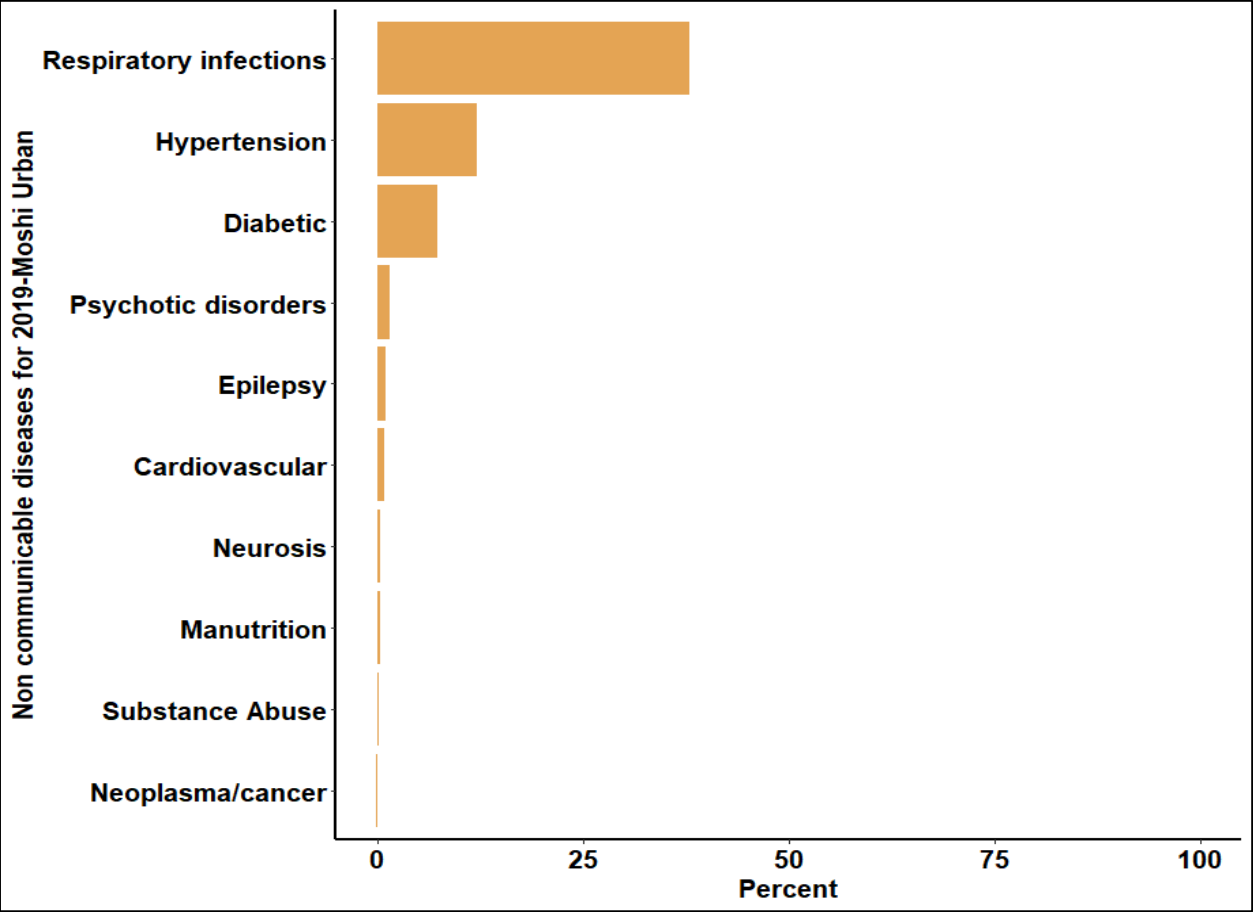


Communicable diseases

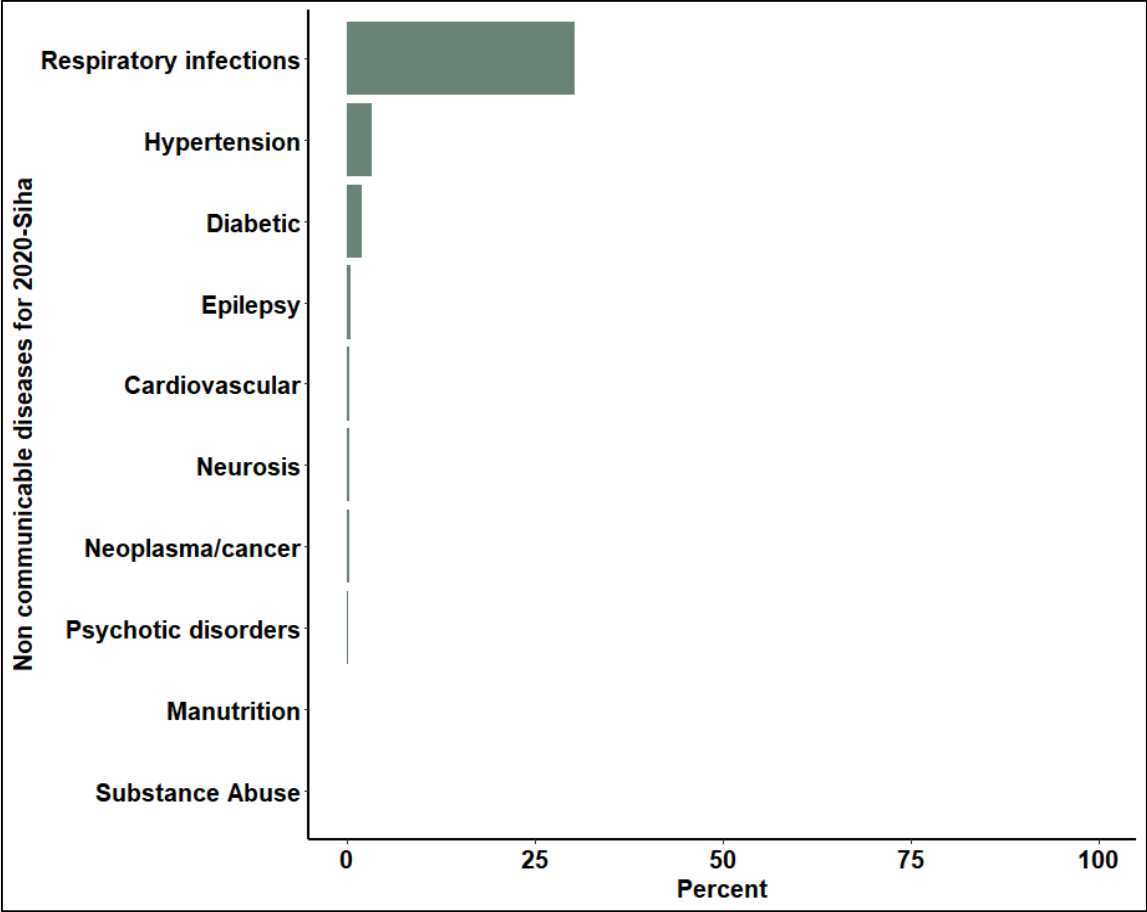


Non communicable diseases

Moshi Urban



Siha



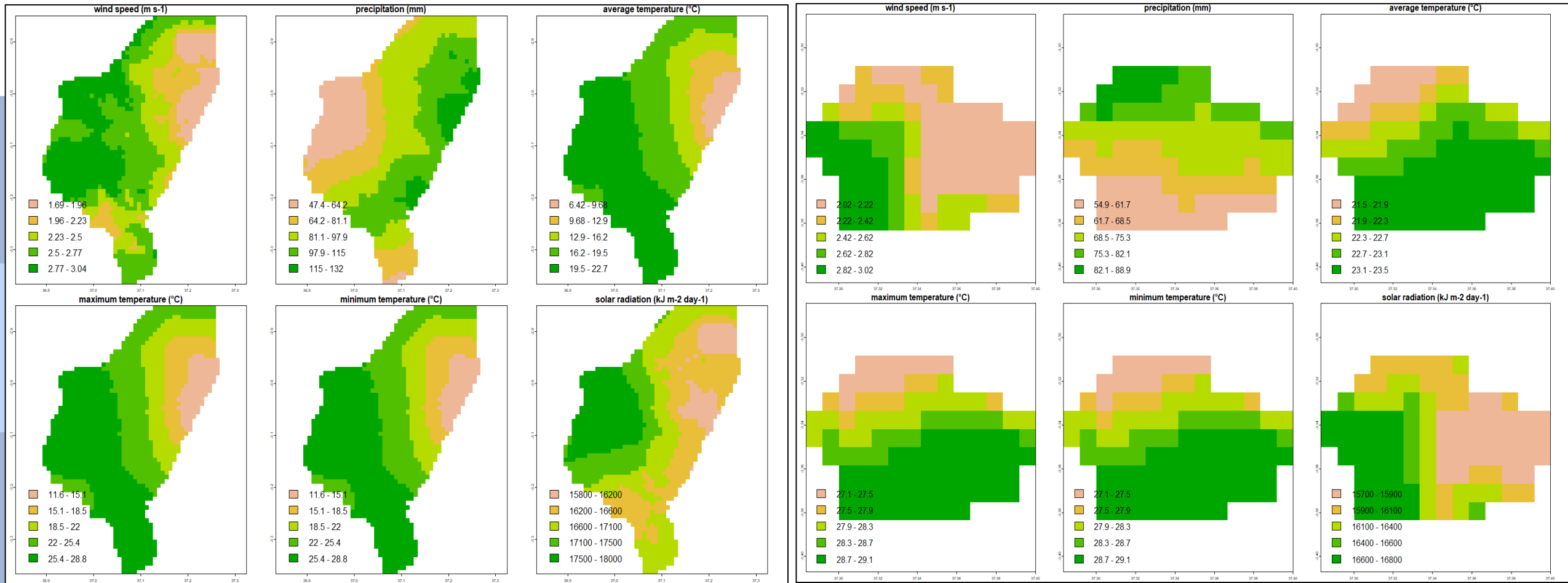
Environmental factors undertaken in the study

- Altitude
- Greenness
- Avg no. of days with rainfall /per year
- Avg no. of days with sunshine/per year
- Mean temperature/per year
- Seasonal temperature
- Soil microbe for Kilimanjaro
- Distance from water bodies /main source of water
- Air quality index
- Wind direction
- Wind speed
- Main crops grown
- Land use
- Size of agricultural plots
- No. of industries per area
- No. of small scale industries
- No. of large scale industries
- No. of gas stations in the district
- No. of main roads
- Traffic capacity per area

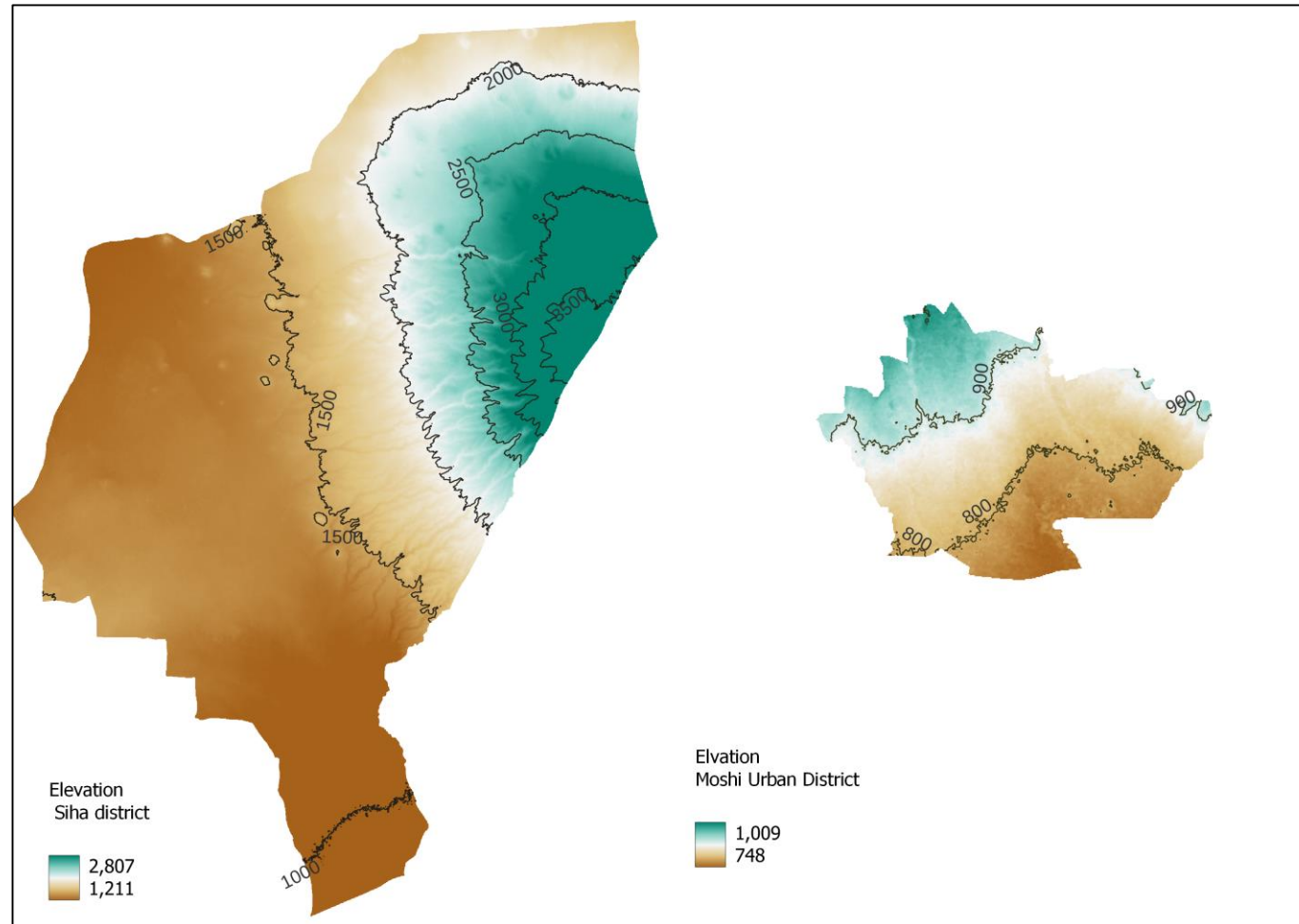
Climatic variables

Siha

Moshi Urban

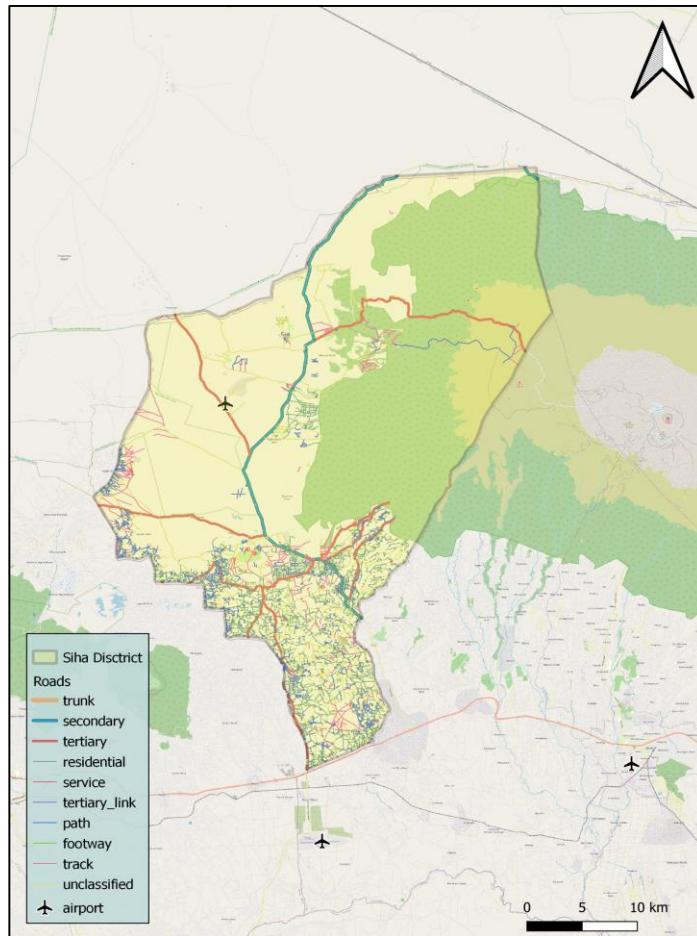


Topographic variables

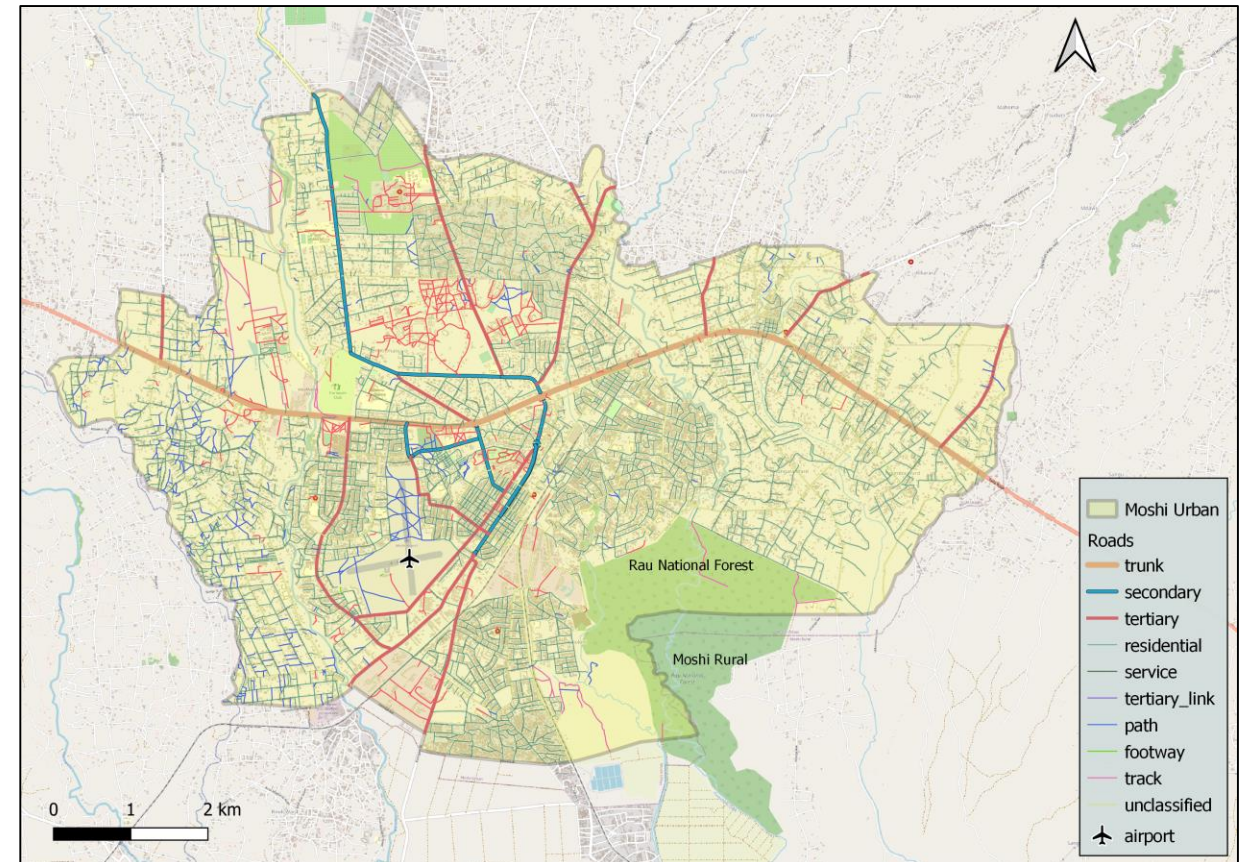


Road Network

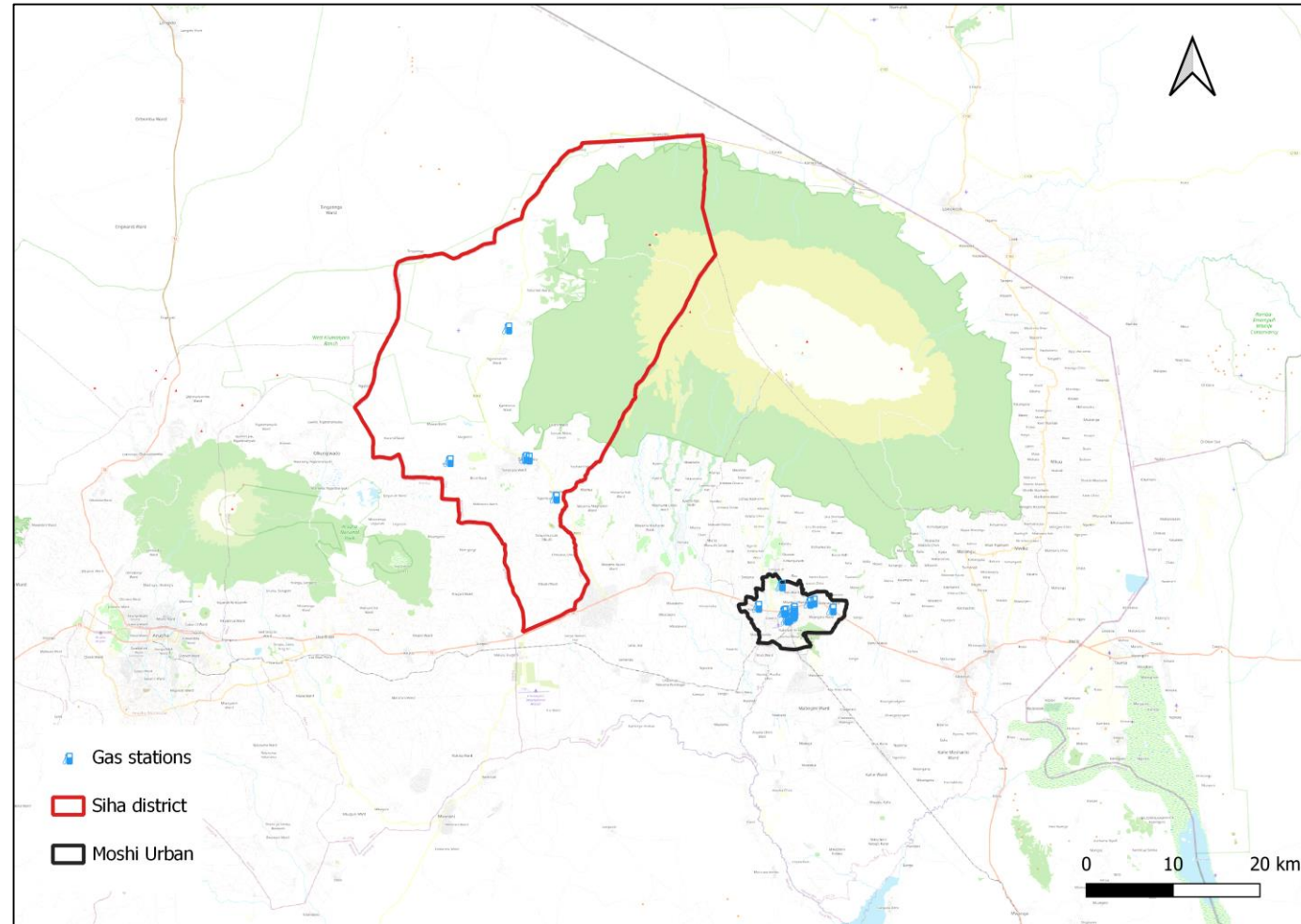
Siha District



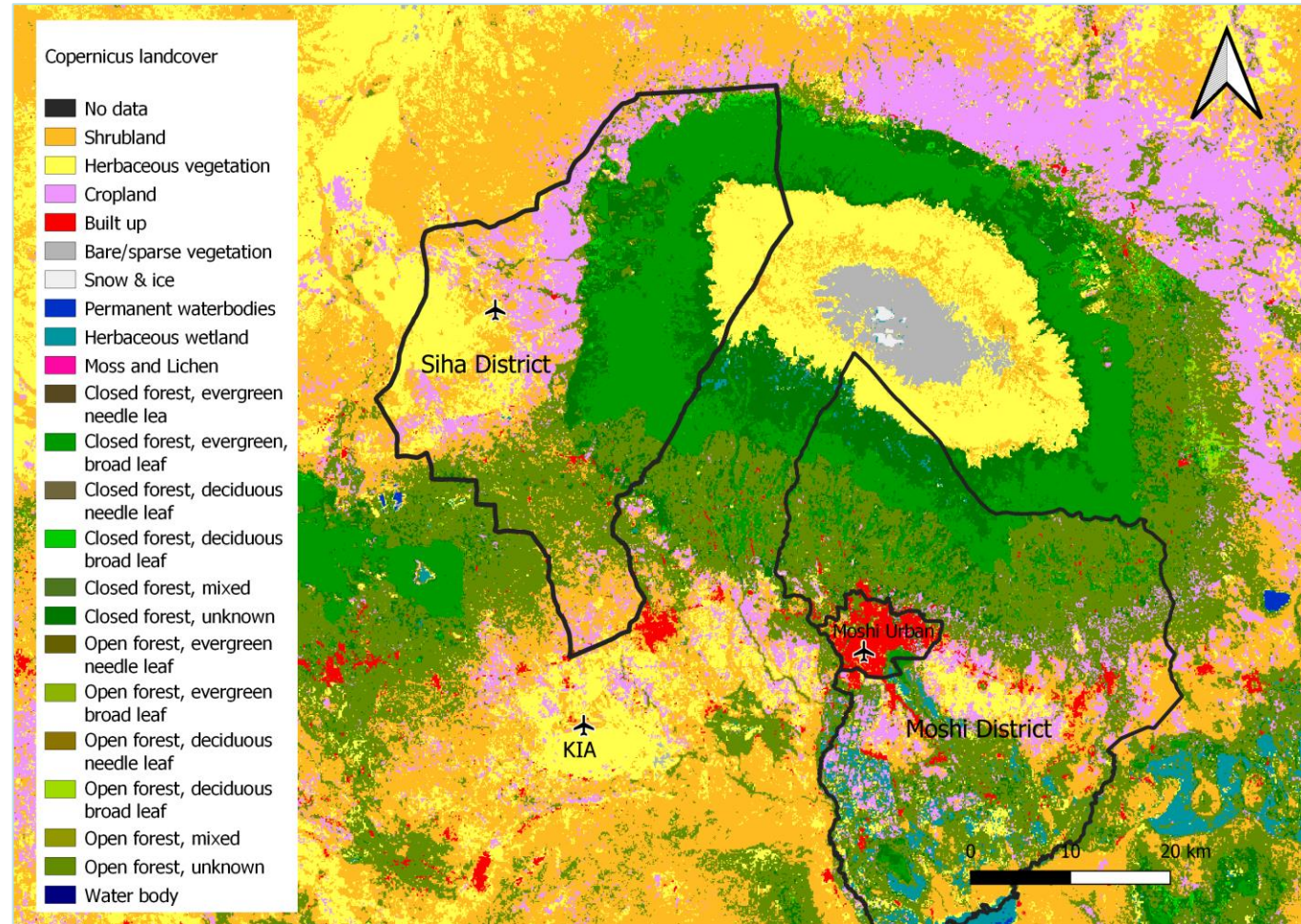
Moshi Urban



Gas stations

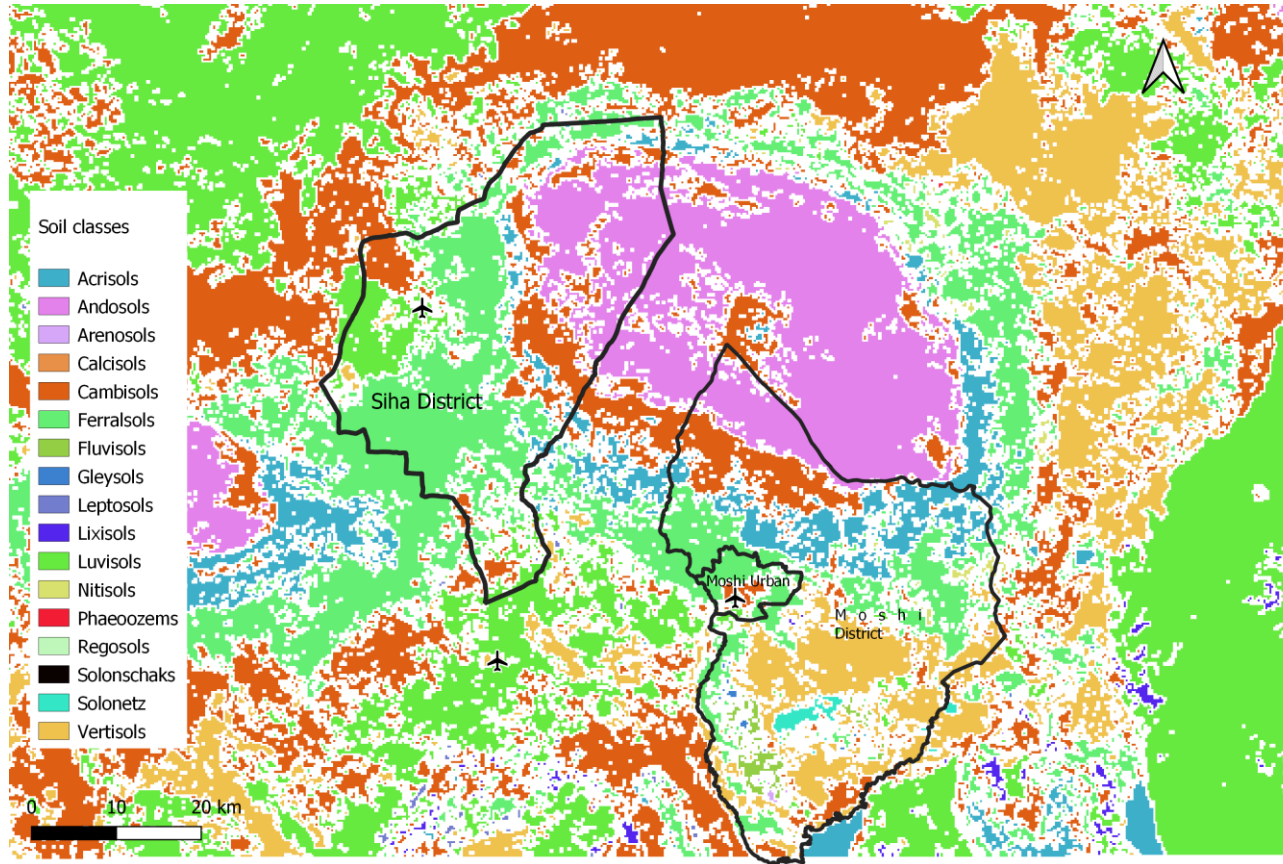


Land use



Copernicus data (100 m, 2019)

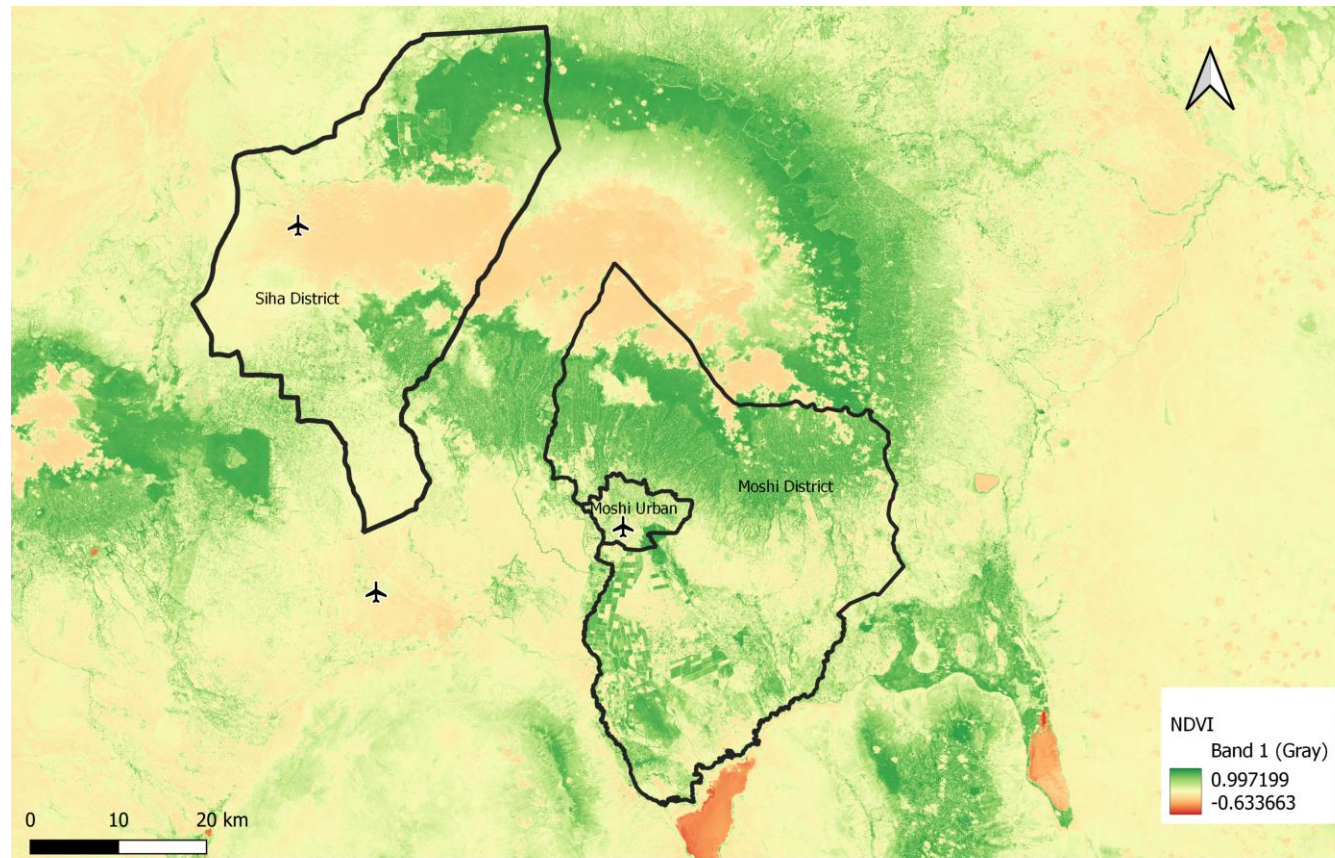
Soil grids data (250 m resolution)



Dominant soil classes

1. Cambisols
2. Andosols
3. Ferralsols
4. Vertisols
5. Acrisols

NDVI – To measure greenness (10 m resolution)



$$NDVI = \frac{NIR - Red}{NIR + Red}$$

Air quality and wind direction

- To be collected from Tanzanian Meteorological Agency

Potential statistical analysis

- Chi square test
- Mixed effect models

WP2

Objectives

- assessing the epidemiological characteristics of asthmatic patients,
- comparing rural and urban asthmatic patients,
- elucidating associations of living conditions on asthma symptoms, in order to
- better understand the environmental drivers of Asthma pathogenesis

Status

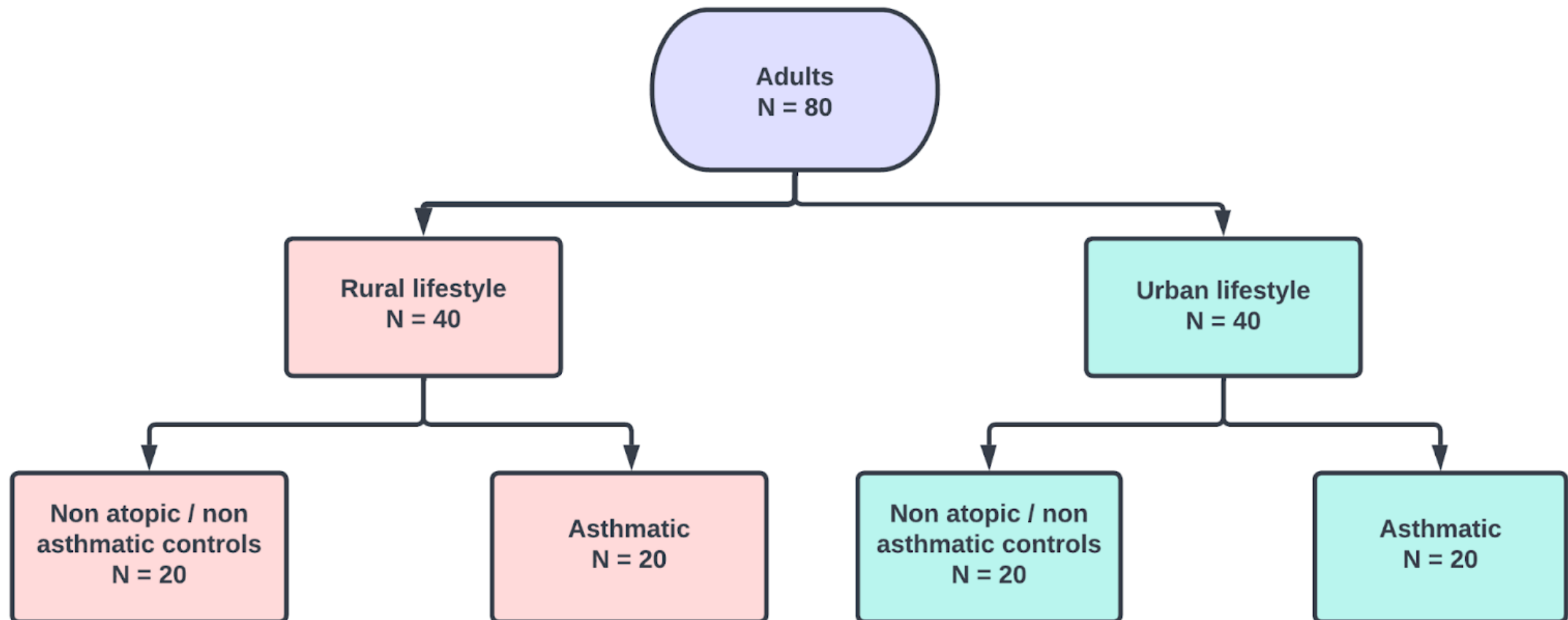
Years in focus – 2018, 2019, 2020 (and 2021)

Patient data collection – to be collected (August-September 2022)

Environmental variables - Ongoing

Methods and analysis – Ongoing

Data collection



POSSIBILITIES

Finding relationships between environmental factors and the incidence of a given (non-communicable) disease.



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July 2021



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Article Contents

Introduction

EPIDEMIOLOGY / HEALTH SERVICES RESEARCH | MAY 05 2021

Land Cover of Early-Life Environment Modulates the Risk of Type 1 Diabetes

FREE

Noora Nurminen; Damiano Cerrone; Jussi Lehtonen; Anirudra Parajuli; Marja Roslund; Maria Lönnrot; Jorma Ilonen; Jorma Toppari; Riitta Veijola; Mikael Knip; Juho Rajaniemi; Olli H. Laitinen; Aki Sinkkonen; Heikki Hyöty

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Corresponding author: Heikki Hyöty, heikki.hyoty@tuni.fi

Diabetes Care 2021;44(7):1506–1514

<https://doi.org/10.2337/dc20-1719> [Article history](#)

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OBJECTIVE

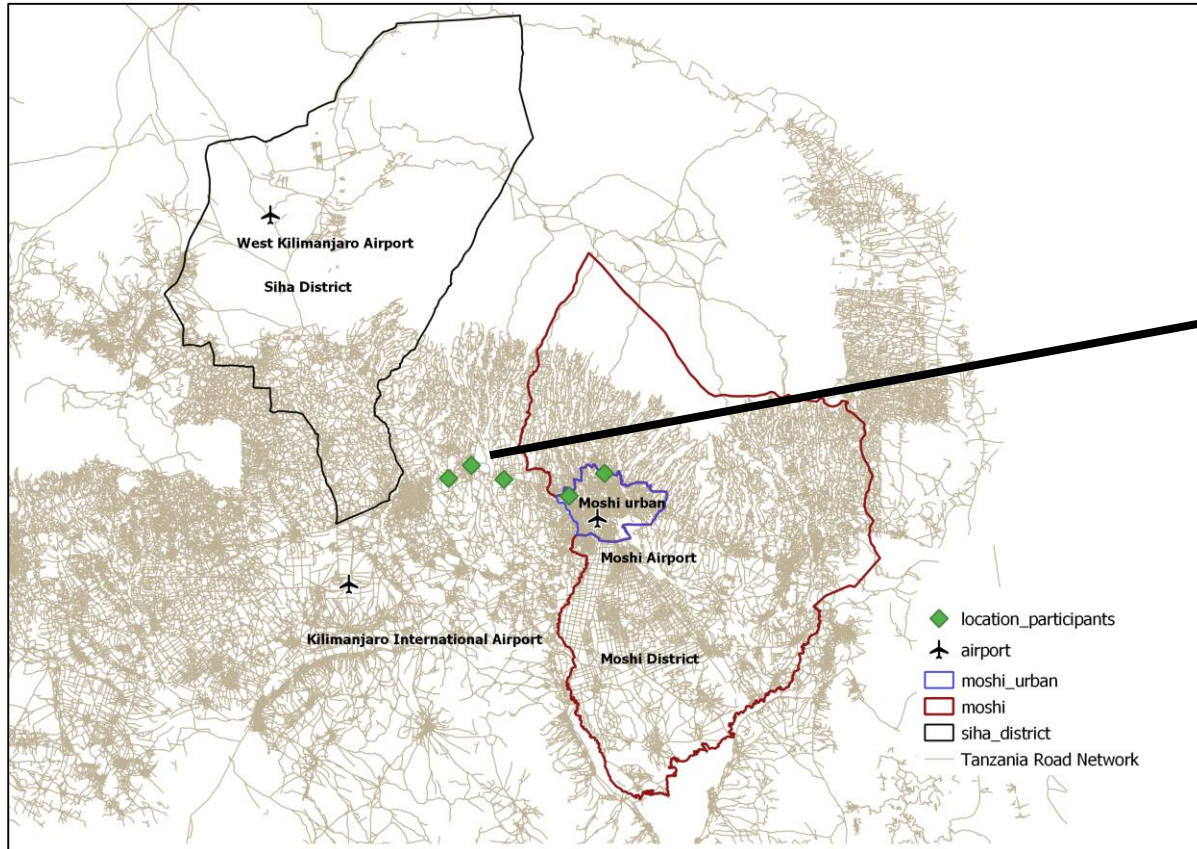
Environmental microbial exposures have been implicated to protect against immune-mediated diseases such as type 1 diabetes. Our objective was to study the association of land cover around the early-life dwelling with the development of islet autoimmunity and type 1 diabetes to evaluate the role of environmental microbial biodiversity in the pathogenesis.

32

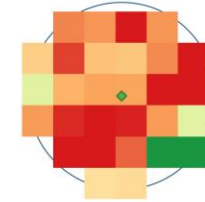
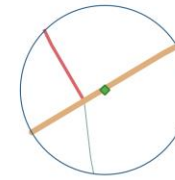
Philipps  Universität Marburg

Kilimanjaro and Marburg | 14.07.2022

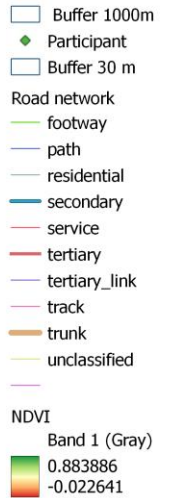
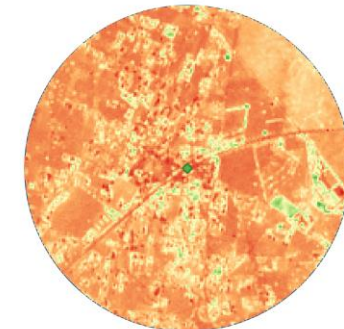
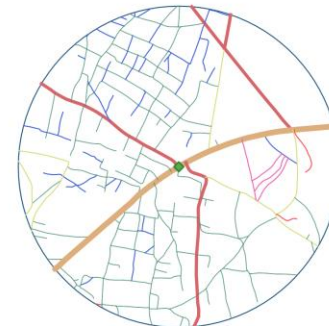
Analysis using spatial buffers



Buffer with 30 m radius

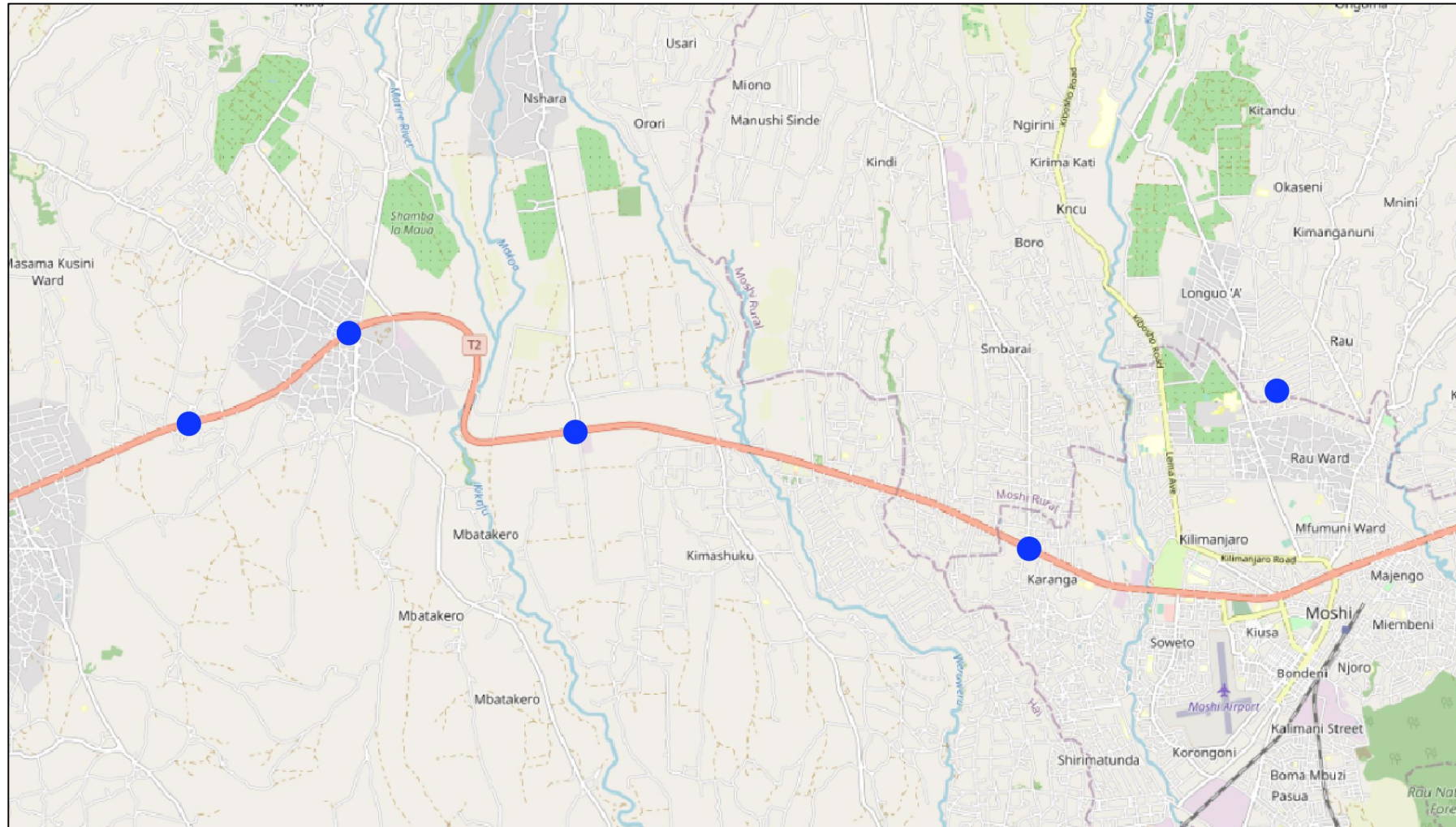


Buffer with 1000 m radius

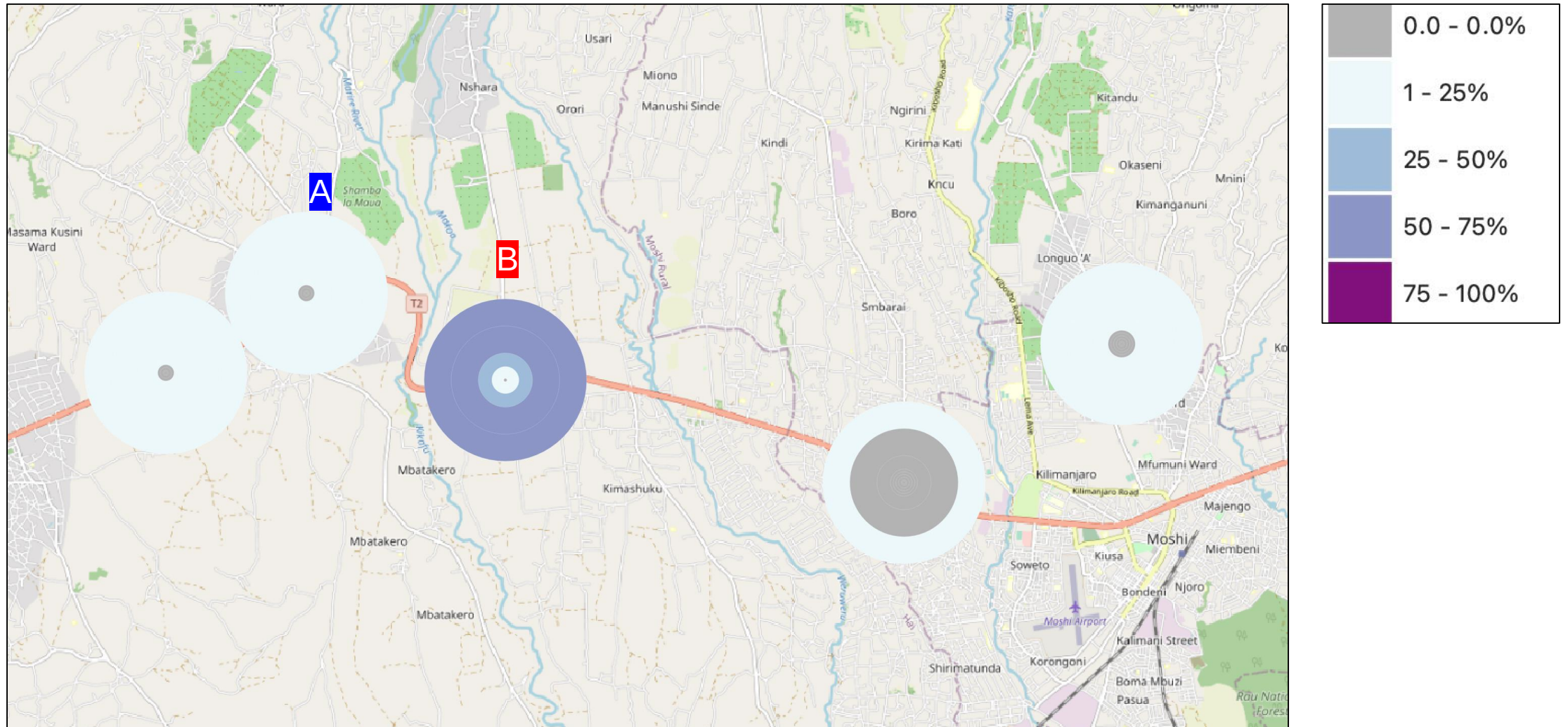


Selected buffers – 30m, 50m, 100m, 150m, 200m, 250m, 500m, 1000m and 1500m

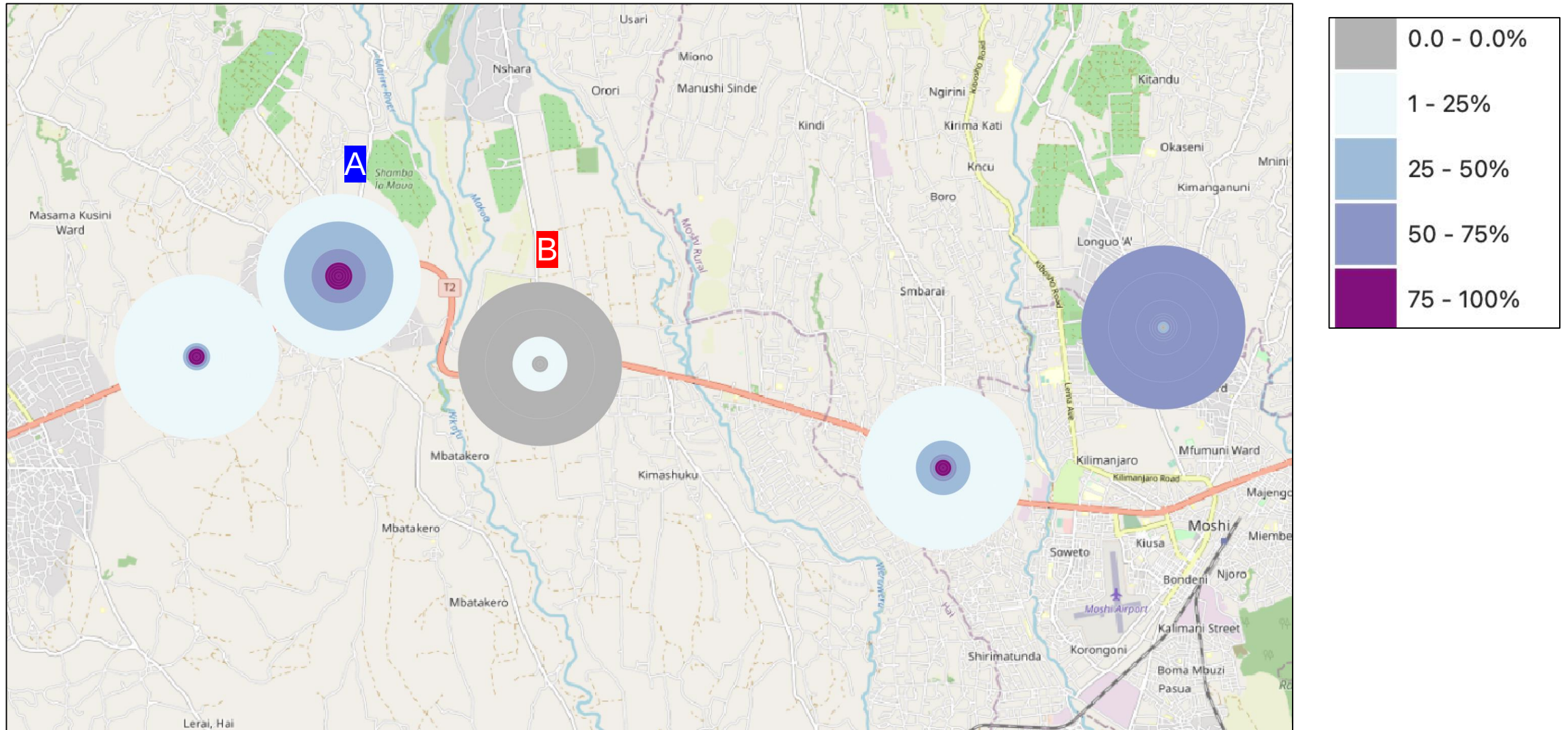
Demo land-use analysis with sample data



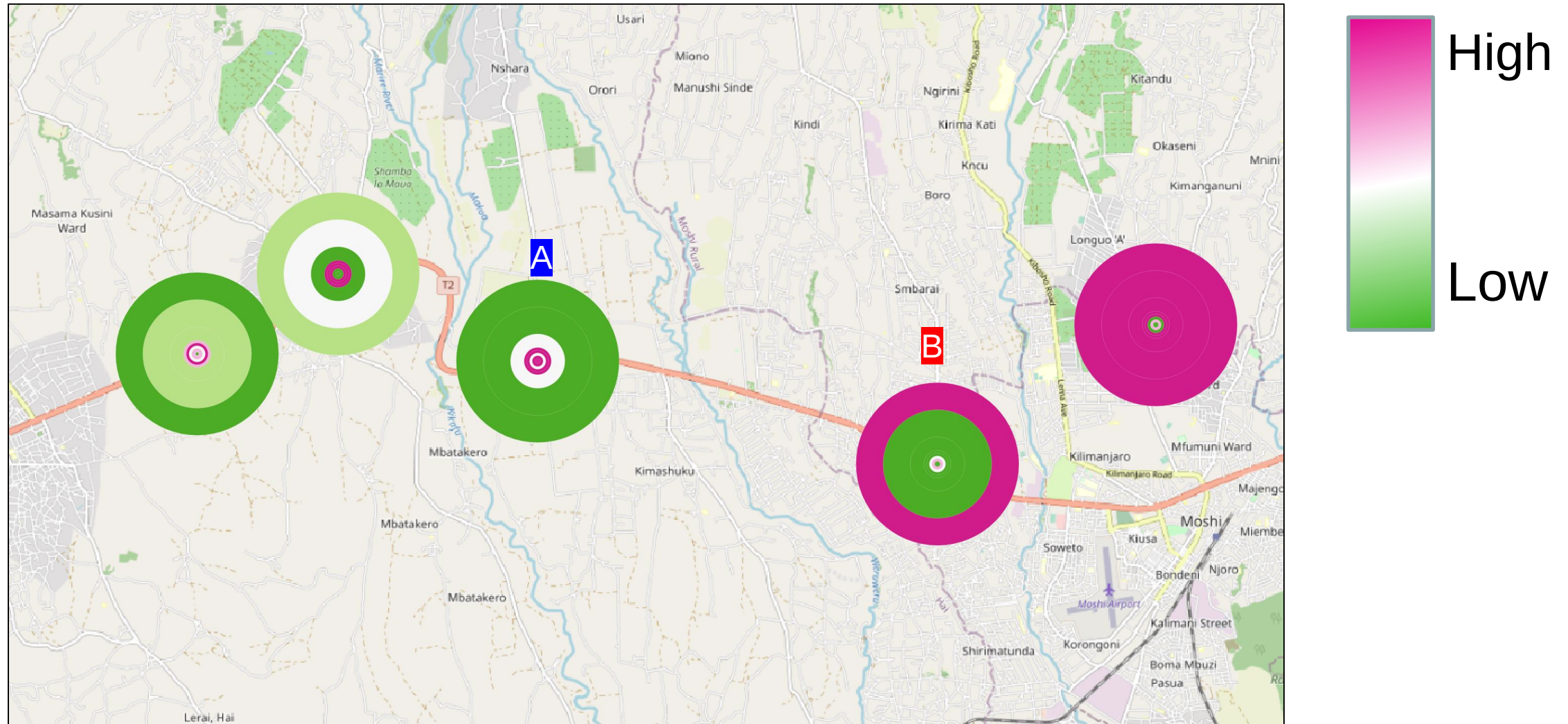
% of Agricultural land within a given radius



% of Urban land within a given radius

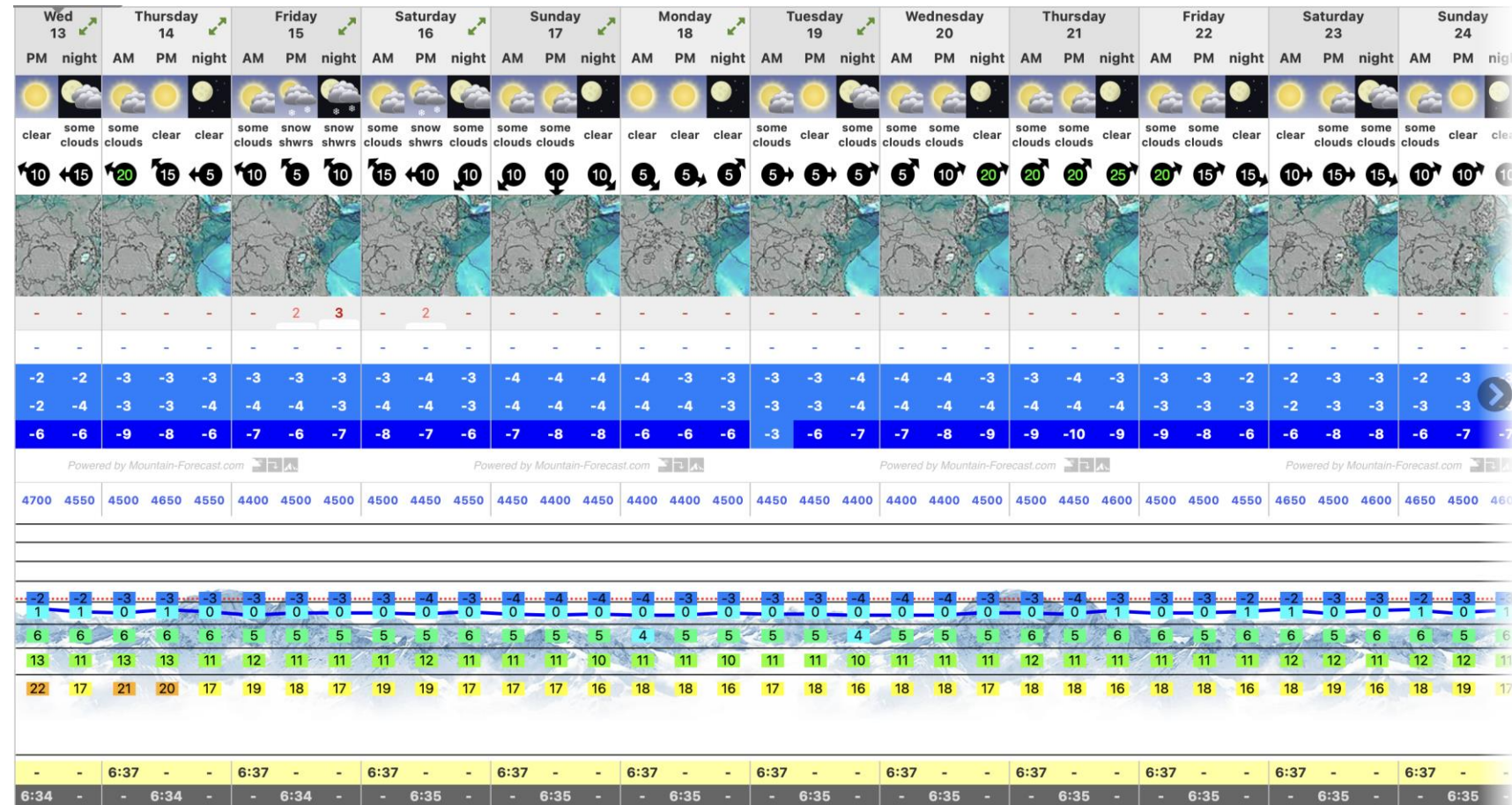


Diversity of land uses (Shannon-Weiner index)



Association analysis through environmental interaction

- Macro-weather conditions
- Localized-weather conditions (pressure, rainfall, humidity)
- Wind direction (transporting materials, dust, pollen etc.)



Our role: Understanding the mechanisms of developing the disease and finding spatial predictors

Discussion

- Detail the mechanisms
 - Define predicting variables
 - What are the working assumptions? Why and what environmental factors may increase the incidence?
- Data quality
 - Verifying the quality and consistency of the medical data captured during the study
 - Assuring correct data classification
 - Define temporal limits (when in life we expect the development of the disease)
- Spurious correlations
 - Using wrong variables
 - Wrongly translating the mechanism into spatial indicators (Eg. At what distance an environmental factor would affect the subject. At what distance a forest from a person's home, has an effect on the disease mechanism?)
 - Unknown variables
- How would an ideal dataset look like